Zixun Zhou 2026年06月16日

ZIXUN - Realise Blog

INSTRUCTION

🥚HOW TO READ THIS PADLET🥚

🥚 FIELD 1 — TYPE
what kind of card this is (question / insight / making / context / primary / precedent).

🥚 FIELD 2 — DATE
when the thinking or the build happened, so the trajectory stays visible.

🥚 FIELD 3 — ID
where it sits in the chain. Prefix `RE-` = Realise. Letter after the hyphen tells the thread:
- `RE-Q` — the question spine (read Q1→Q6 in order)
- `RE-I` — reasoning, pivots, honest boundaries
- `RE-M` — hands-on build
- `RE-S` — overview / how-to-read / maps
- `RE-P` — first-hand demo evidence
- `RE-C` — precedent

🥚 FIELD 4 — STATUS
LOAD-BEARING = the argument stands on this.
SUPPORTING = serves the line.
RUNNING = unfinished / uncalibrated / paused on purpose. DONE = finished build kept as process evidence.

If you only follow one thread, follow **RE-Q1 → RE-Q6**. That is the argument Realise hangs on.
Cross-references back into Investigate use `AL-` IDs (especially `AL-Q7`, `AL-I19`, `AL-P0`–`P4`).

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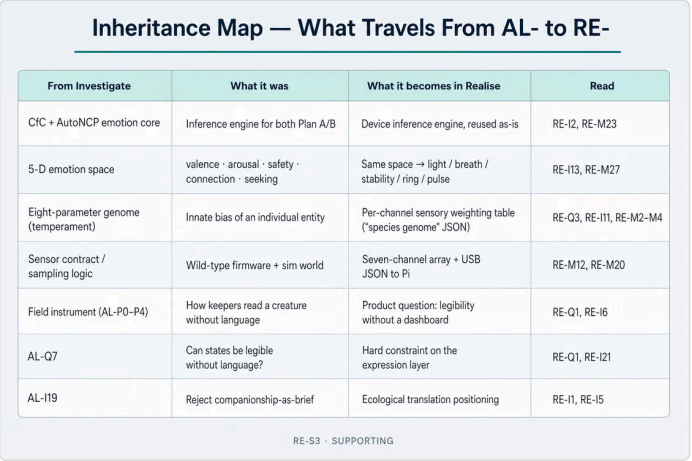
Inheritance Map — What Travels From AL to RE

Realise is not designed from a blank page.
This card is the handoff list — what moves,
what changes shape, and where to read it.
What does **not** travel unchanged: the wild-
type's simulated room, the nurtured-type's
caregiver LLM, and any claim that five weeks
of deployment equals long ecological learning.

DATE
2026.7.1

ID
RE-S3

STATUS
SUPPORTING



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版块 4

AI Life Project Overview: From Investigate to Realise

Research Origin

This project began as an inquiry within Artificial Life (ALife): if an AI entity is given physical sensors, continuous memory, and the passage of time, will it develop **emergent behaviours that were never explicitly designed into it**? The question deliberately adopts ALife's definitions of "internal state" rather than philosophical definitions of

consciousness — the project is concerned with whether behaviour and state can grow out of the interaction between architecture and experience, not with answering the unanswerable question of whether an AI "has" consciousness.

Investigate Phase: *Three Parallel Lines of Research*
Technical experimentation: the Plan A / Plan B twin-track

- Plan A (Wild Type) is an entity with no language model — a randomly initialised CfC neural network driven purely by real physical sensors — exploring what a cognitive entity becomes with no socialised input whatsoever.
- Plan B (Nurtured Type) instead has a large language model acting simultaneously as social environment and world simulator, raising the entity within a specifically defined apartment setting with caregiver characters.

The comparison across the two tracks addresses a central question: **what does social experience actually shape, in a cognitive system built from nothing?**

The emotional core uses a CfC (Closed-form Continuous-time) + AutoNCP neural circuit architecture. A five-dimensional emotion space (valence, arousal, safety, connection, seeking) is driven by a prediction engine grounded in interoceptive inference, alongside an eight-parameter genome system rooted in neuroscience (Panksepp's affective theory, polyvagal theory) that defines each individual entity's temperamental tendencies.

Theoretical research
Drawing on ALife literature (e.g. Nisioti et al. 2024 on the intersection of LLMs and ALife), attachment theory from developmental psychology (Bowlby), and affective neuroscience, this line of work builds an academic foundation for the project's central claim of "emergent behaviour," so the project does not remain purely a technical demonstration.

Field research
A structured interview series (six core questions, roughly ten minutes per participant) was conducted with people who care for small living creatures. The central probe: without language, how does a person first notice that a living creature's state has changed — was the first sign something seen, heard, or felt, or was it simply sensed? The interview deliberately included a speculative question: describing to participants a digital being that starts blank, senses the real world through physical sensors, has no language, and slowly grows its own personality — then asking for their gut reaction and what conditions would make them start to care about it. This interview design directly foreshadowed the pivot in the Realise-stage product direction.

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The Pivot: From "Emotional Companionship" to "Ecological Translation"
The project's initial output direction leaned toward an emotional-companionship device — something in the territory of an electronic pet or Tamagotchi. This positioning had two problems: it overlapped thematically with the Unit 2 ADHD therapeutic toy project, and it had limited interdisciplinary reach; more importantly, it failed to genuinely respond to the core insight accumulated through Investigate — the field interviews were never really asking "how does a person come to feel affection for a thing," but rather **how a person comes to understand, intuitively and without a digital dashboard, what a living system is currently experiencing.**

The corrected positioning is this: the device senses the environment itself — soil, water, air — not the emotional needs of the user. By "inhabiting" different species genomes, the entity translates environmental states that humans cannot directly perceive (soil moisture deficit, risk of water-body hypoxia, poor indoor air quality) into an emotional, biological form of physical expression — light, vibration, sound — so that people can understand the health of an ecosystem through intuition rather than data. The sense of companionship is a natural by-product of this mechanism running, not the design goal — and this distinction sets a hard line for the entire expression-layer design: the output communicates what the environment is experiencing, not an attempt to please the user.

Realise Phase: How the Device Grows Directly Out of Investigate's Technical Base
The Realise output is not a system designed from scratch — it extends the core architecture already built and validated during Investigate into a new physical form:

- The CfC + AutoNCP emotional core from Plan A/B is reused as-is as the device's inference engine
- The eight-parameter genome system that originally governed each entity's "temperament" is extended into a per-sensor-channel weight vector, allowing the same hardware to "inhabit" different ecological niches (soil/root system, freshwater, indoor air) simply by switching a "species genome" JSON configuration file — this is the engineering realisation of the "chameleon" design concept

- A two-layer sensory weighting transform (comfort curve + genome weight) translates raw readings from a seven-channel universal sensor array (temperature, moisture, gas composition, light, sound, pH, electrical conductivity) into a weighted signal fed to the CfC core
- Rather than relying on hard-coded, literature-estimated thresholds, comfort ranges are auto-calibrated from observation data gathered during the device's initial deployment — this design both addresses the question of where the thresholds come from in a verifiable way, and naturally maps onto the entity's existing developmental narrative (a newborn first becomes familiar with its environment before forming judgements)

Physically, the device takes the form of a biomimetic, egg-shaped enclosure housing a two-tier Raspberry Pi + Arduino compute architecture, a rechargeable battery, and a ring of 12 RGB LEDs as the primary channel of emotional expression. Probe-style sensors insert into soil or water; the enclosure is realised through 3D printing.

❗❗**Honest Boundaries of Scope**❗❗
The following are limitations the project has actively flagged through self-examination, rather than issues raised after the fact:

- Within the five-week Realise window（longest）, what can credibly be demonstrated are short-timescale, induced events (e.g. artificially triggered drought or a sudden temperature shift), not the slow ecological trends that take weeks or months to observe (e.g. soil acidification) — the latter is left as a direction for future long-term deployment
- The emotional response pattern to environmental stress currently relies on designed sensory-weighting rules, not behaviour learned by the network through sustained experience — the project maintains a clear distinction between the two, and architecturally separates "weighting rules" from "the CfC core's emergent temporal dynamics," to avoid misrepresenting designed behaviour as emergent behaviour
- A single-point sensor reading can only represent the state at the probe's specific location, not the state of an entire environment — this spatial limitation is explicitly acknowledged within the design narrative rather than left implicit

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The Jump Was Not Inspiration — It Was Written in the Wild Type's Own Conclusion

People keep asking me how I got from a neural network experiment to making a physical object, as if there were a moment of inspiration. There was not. There was a methodological dead end, and the object is the only way out of it.

Here is what actually happened.

The wild type ran for two virtual years and produced a headline number that looked like success. Underneath, it had collapsed into a single internal state and never left it. Three of its five emotional outputs had saturated and stopped carrying information. The most common sequence in the entire log was state six, followed by state six, followed by state six — forty-four thousand classified moments, all in the same group.

And my own scoring measure gave that its highest marks for stability. I had built the measure to detect stable patterns, and it faithfully detected the most stable pattern available to it, which was death. A metric that awards top marks to a system that has stopped is not measuring what I claimed it was measuring.

That was bad, and it was fixable. The thing that was not fixable is what I wrote down next, and it is the sentence this whole Realise phase grew out of:

A world I wrote to be meaningless is still a world I wrote.

I chose how long the day was. I chose how often a visitor came, how loud the room was, how much everything fluctuated. So when nothing structured emerged, I had no way — none, in principle, not just in practice — to separate "the creature failed to find structure" from "there was no structure in my invented world to find." Both explanations produce exactly the result I got. From inside a simulation I cannot tell them apart, and no amount of longer runs or better metrics fixes that, because the problem is not in the measurement. It is in the fact that I am both the experimenter and the author of the universe.

There is exactly one move that escapes it. Put it in a real room. A real room is the only world I cannot be accused of having authored.

So the leap to a physical object was not a change of subject and it was not a design ambition arriving from outside. It is the first item on the wild type's own list of what to do next, written in the conclusion of that experiment before Realise existed as a phase. Everything downstream follows from it — the sensors in RE-

I30 exist because a real room has to be touched; the honesty rules in RE-M12 and RE-M24 exist because a real room, unlike a simulation, is allowed to be silent, broken, or ambiguous, and my simulator never was.

Then a second thing happened, and this is the part that is genuinely a leap rather than a forced move.

Once I accepted that the entity needed a real environment, I had to choose which real environment. My instinct was a room with a person in it, because that is what the simulation had been — someone walking past every five to forty minutes. But the field interviews were sitting on my desk, and what they were about was not people bonding with creatures. They were about people reading creatures: noticing without language, mistranslating, catching it late, being frightened of damaging something once it had become singular.

Which reframed the whole thing. I had been building a system to see whether a machine could develop readable states. My participants were describing the same problem from the other side — how a human learns to read a state that has no words attached. And the environments in their answers, soil and water and air, were full of states that are exactly like that: real, consequential, and completely invisible until they are an emergency.

So the object stopped being a creature that needed a real room and became a translator that needed a real medium. That is where RE-I1's pivot actually comes from, and it is why RE-I6 says the field work forced the product question rather than validating a product I had already chosen.

Two honest notes on this card. The wild type's own to-do list said to rebuild that same six-sensor board and rerun the experiment in a real room. I have not done that. I built something different instead — a seven-channel ecological device with a different sensor list and a different purpose, which is not the same experiment and does not settle the question that experiment failed to settle. And the second: a longer run of identical code, without the stopping rule I had imposed, showed a live coupling between physical presence and the connection dimension by six virtual years. My stopping rule ended the experiment just before the only interesting thing in it began. That is worth remembering every time this Padlet talks about what five weeks can demonstrate.

TYPE 🟨 INSIGHT

DATE 2026.6.20
ID RE-I31
Status LOAD-BEARING

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What This Phase Is Not
Three refusals, stated up front so nothing later can quietly claim them.

Not a second experiment.
Investigate already ran the twin-track (nurtured / wild). Realise does not re-ask “will behaviour emerge?” as if that work had not happened. It asks a different question: can the architecture that survived Investigate become a physical translation medium for an environment a person cannot feel directly?

Not an electronic pet.
The first product instinct was companionship — Tamagotchi territory. That path was rejected in Investigate (`AL-I19`) and is rejected again here. The object does not exist to be loved. It exists to make an ecosystem’s state legible. If companionship appears, it is a by-product of that mechanism, not the brief (`RE-I1`, `RE-I5`).

Not a claim that the network “learned ecology.”
Within this window, stress responses ride on designed sensory-weighting rules. Those rules are not the same thing as behaviour earned through long experience. The architecture keeps them separable on purpose (`RE-Q6`, `RE-I3`, `RE-I22`). Misrepresenting design as emergence would be the easiest way to make the project look stronger than it is — so this card forbids it.

DATE
2026.7.1

ID
RE-I0

STATUS
LOAD-BEARING

0

Positioning Statement

The long-term vision is a general-purpose device for the emotional translation of ecological environments — the same hardware, switching between species genomes to inhabit different ecological niches, translating environmental states that humans cannot directly perceive into an emotional, biological form of expression, so that people can understand what an ecosystem is experiencing through intuition rather than a data dashboard. The twin-track experiments and field research of the Investigate phase, together with the device prototype of the Realise phase, form a continuous inquiry into a single underlying question: can a system that persists continuously through time, shaped by sensory experience, help humans relearn how to understand — intuitively rather than numerically — a living world they cannot directly perceive?

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Why This Is Not Another Wellbeing Toy

Unit 2 already occupied “emotional wellbeing object” territory. A companionship egg would have been a thematic remake with better lights.
Realise has to sit in a different neighbourhood: ambient / environmental sensing that tries to make invisible conditions graspable without forcing the user through a spreadsheet — more “legibility of a world that is not you” than “an object that soothes you.” I am not loading a fake bibliography onto this card. When a specific exhibition, device, or paper becomes load-bearing for the argument.
What I refuse from the pet neighbourhood: bonding loops as success metrics.
If a reader can swap ANIMA for a generic ADHD soothe-toy and the story still works, the positioning failed — go back to `RE-I1` and `RE-I6`.

TYPE 🟨 INSIGHT
DATE week 5
ID RE-I29
Status SUPPORTING

0

版块 1

What must stay legible without a dashboard?

This is the Realise form of the problem Investigate already named (`AL-Q7`, and the earlier placeholder `RE-01` inside the field cards). If the only way to understand the object is to open an app and read numbers, the project has failed its own brief. The field interviews were never really asking whether people would *like* a digital creature. They asked how a person first notices that a living system’s state has changed — seen, heard, smelled, touched, or “just felt” — and what it costs to learn that reading (`AL-P0`–`P4`). So the constraint for Realise is concrete: - The primary output is physical and non-verbal (light, breath, optionally extreme tone — never speech). - A phone journal or PWA may exist as a secondary record for the portfolio and for debugging. - Evaluation criterion: can a person standing next to it form a usable intuition about *what the environment is going through*, without being handed a chart? Everything in the expression layer has to answer this question first. Charm is irrelevant. Clarity under stress is not.

DATE
2026.7.1

ID
RE-Q1

STATUS
LOAD-BEARING

0

If the entity inhabits an ecosystem, whose feelings are being shown?

There are two easy mistakes, and both produce a product that looks like Unit 2 again.

Mistake 1 — show the user's feelings.
A device that brightens when you approach, soothes when you are sad, performs caretaking. That is companionship design. It centres the human emotional need.

Mistake 2 — invent a pet's feelings and ask the user to love them.
Same territory, different costume.

Realise chooses a third centre: **the environment's stress, read through a species-shaped sensory weighting.** When the light turns urgent, it should mean *this soil / water / air is departing its own recent rhythm* — not *the object wants you*. The entity is a translation medium. It "inhabits" a genome the way an actor inhabits a role: the performance is about the world being sensed, not about winning attachment.

This sounds semantic until it hits the LED map. Every mapping decision after this card (`RE-I5`, `RE-I13`) is judged by one test: does the output report the environment, or audition for affection?

DATE
week5

ID
RE-Q2

STATUS
LOAD-BEARING

0

Can one body wear many natures?

Investigate's genome was temperament — eight parameters that biased how an individual entity felt. Realise keeps the word and changes the job.

Here a genome is **not different hardware**. It is a sensory weighting table: for each of the seven channels, a weight (0–1), a comfort range, and a sensitivity curve. Switch the JSON, and the same egg, the same probes, the same CfC core inhabit a different niche — soil/root, freshwater, indoor air.

Weight zero does not mean the sensor is off. The channel is still recorded. Zero only means *this nature does not care about that dimension* (`RE-I10`, `RE-I11`). That distinction matters later: a muted channel can still teach the system that the medium has changed, or that the wrong genome is loaded.

The "chameleon" claim stands or falls on this card. If I need a different board for water than for soil, I do not have one body wearing many natures — I have a product line.

DATE
week5

ID
RE-Q3

STATUS
LOAD-BEARING

0

Where do the thresholds come from — literature, or lived observation?

Hard-coded comfort ranges copied from a paper are convenient and weakly defensible. They also pretend the device arrived knowing what "normal" is for **this** patch of soil, **this** tank, **this** room.

Realise's preferred answer: **auto-calibrate from observation during an initial deployment window.** The entity first becomes familiar with its environment's own recent rhythm, then flags departures from that rhythm. That is closer to interoceptive / predictive logic already used in Investigate, and it maps cleanly onto a developmental metaphor without pretending the metaphor is biology.

This does not make thresholds "objective." It makes their origin local and inspectable. Anyone reading the work can ask: calibrated on what days, under what interventions, with which genome loaded?

Literature still informs the *shape* of the curves and which channel combinations count as signature stress (drought, hypoxia risk, poor ventilation). Lived observation supplies the *centre* of the comfort band. Confusing those two sources is how a demo becomes a fiction.

DATE
week5

ID
RE-Q4

STATUS
LOAD-BEARING

0

What can five weeks honestly demonstrate?

A slow ecological trend — soil acidifying over months — cannot be "shown" inside a Realise window without lying with time. So the credible demo class is narrow on purpose:

In scope: short-timescale, induced events. Withhold water until moisture and coupled channels move. Shift temperature abruptly. Switch genome mid-session and show the same hardware re-weighting the same world.

Out of scope for proof: week-scale chemical drift, "the network learned a new ecological concept," claims about whole-environment health from one probe tip.

This card exists so later demo cards (`RE-P1`–`P3`) cannot quietly inflate. The boundary is also written as insight in `RE-I3`. Q5 is the question form; I3 is the sworn statement.

Five weeks can still be enough — if what is demonstrated is *translation under induced stress*, not *mastery of ecology*.

DATE
week5

ID
RE-Q5

STATUS
LOAD-BEARING

0

When light breathes, is that emergence or design?

Viewers will see breathing light and say "it feels alive." That sentence is dangerous if I let it mean "the liquid network invented this behaviour."

Split the stack:

1. **Designed layer** — sensory weighting rules, genome tables, LED mapping (hue ← valence, breath rate ← arousal, tremor ← safety, and so on). Authored. Tunable. Not earned by experience inside the five-week window.
2. **Dynamic layer** — CfC continuous-time response to the weighted stream: how the internal state moves when inputs change together, how it recovers, how it lags. This is where temporal dynamics can still be interesting without being oversold as "understanding drought."

Realise's honesty rule: **never present (1) as if it were (2).**
Architecture must keep them separable in code and in language (`RE-I22`). If a drought demo looks right because I wrote a moisture weight of 0.9 and a red hue for negative valence, that is successful design — and I should call it design.

Emergence, if any, is a narrower claim about dynamics under those rules — not a synonym for "the lights look emotional."

DATE
week5

ID
RE-Q6`

STATUS
LOAD-BEARING

0

Companionship Is a By-product, Not a Goal

Once `RE-Q2` is answered, the expression layer needs a hard line in plain language.

Goal: translate environmental state into a form a person can read with the body — glance, peripheral vision, gut response — without a dashboard (`RE-Q1`).

Not a goal: make the user fond of the object; mimic pet bonding loops; reward proximity with "happiness"; speak; perform neediness.

Companionship may still happen. People anthropomorphise glowing things. Field participants already reached for words like care, responsibility, and damage when they heard the speculative probe. That reaction is data about *human reading*, not permission to redesign the brief around attachment.

Design test I will reuse on every LED decision:
If I remove the user's desire to be liked from the story, does the behaviour still make sense as a report about soil, water, or air? If no — cut it.

This card is the operational twin of `RE-I1` (the pivot) and `AL-I19` (the rejection). I1 explains why the brief changed. This card explains how that change polices the build.

DATE
week5

ID
RE-I5

STATUS
LOAD-BEARING

0

Field Research Forced the Product Question

The field instrument was never a popularity test for a device I had already decided to build.

`AL-P0`–`P4` asked keepers of small living creatures how they first notice a state change without language — seen, heard, smelled, touched, or "just felt" — and what it costs to learn that reading. The speculative probe (a blank digital being that senses through physical sensors and grows) was there to watch the *human* reaction, not to validate a Tamagotchi brief.

What came back forced the product question sideways. The useful material was not "I would love this." It was material about **reading**: mistranslation loops, delayed recognition, the fear of damaging something once it has become singular, the difference between relating to one individual and to a group. That is a problem of legibility under uncertainty — the same problem `AL-Q7` named for the entity's inner states.

So Realise does not ask: how do I make people attach to an object?
It asks: how does a person understand what a living system is experiencing **without a data dashboard**?

The companionship product failed that question. Ecological translation is the form that can still answer it (`RE-Q1`, `RE-I1`).

DATE
week 5

ID
RE-I6

STATUS
LOAD-BEARING

0

Why Not a Dashboard App Alone

A clean dashboard would be easier to finish in five weeks. Charts for moisture, VOC, pH; thresholds in red. It would also abandon the only constraint that makes Realise continuous with Investigate.

If understanding requires opening a phone and reading numbers, the project has accepted that ecological state is for people who already think in dashboards. The field work pointed the other way: keepers often catch the first signal in the body — a sound, a stillness, a smell, a "just felt" — before they can name a metric.

So the phone may exist as a **secondary** journal and debug surface (portfolio PWA, logs, genome switch UI). It is not the primary interface. The primary interface is the object in the room: light, breath, optional extreme tone — never speech (`RE-Q1`, `RE-I21`).

This is also how Realise stays distinct from a generic IoT plant monitor. The claim is not denser data. The claim is a different channel of understanding.

DATE
week 1

ID
RE-I7

STATUS
LOAD-BEARING

0

Naming the Object — ANIMA / A Living Presence
The name has to carry the corrected brief, not the abandoned one.

ANIMA points at an animating principle — something that makes a body feel inhabited — without promising a pet, a friend, or a therapist. **A Living Presence** describes what you encounter in the room: a continuous presence whose state can be sensed peripherally, not a character waiting to be entertained.

What the name must not do: invite "my baby," "feed me," or "I missed you" as the default story. Those stories pull the expression layer back toward companionship (`RE-I5`).

Naming is a design control. Every caption, portfolio slide, and LED behaviour has to remain compatible with: *translation medium for an environment you cannot feel directly* (`RE-I4`).

DATE

week 5

ID
RE-I8

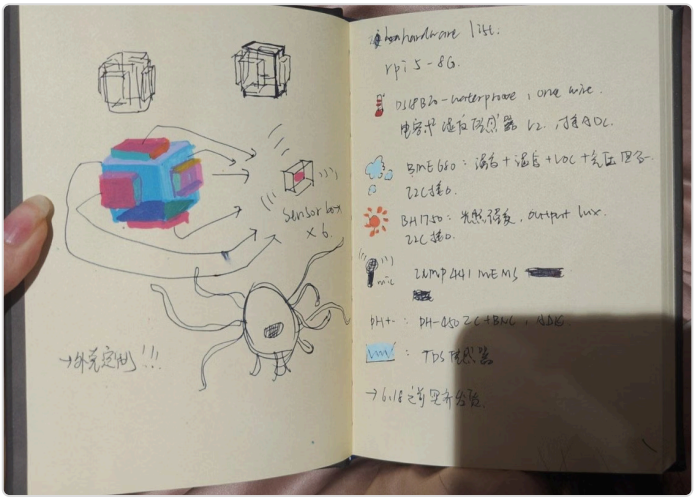
STATUS
SUPPORTING

0

版块 5

APPEARANCE

Why an Egg — I Compared Four Bodies and Killed Three
I did not start with an egg. I started with a clear plastic box, because that is what you build when you are still thinking like an engineer. I put the boards in it, looked at it on my desk, and the honest reaction was: this is lab equipment. Nobody walks past lab equipment and wonders how it is doing. It reads as a machine that has readings inside it, which means the first thing anyone would ask is "so where do I see the numbers" — the exact question RE-Q1 says the project has to survive without.



So I went the other way and sketched a face. Eyes, a soft body, something with ears. It was immediately better to look at. It was also immediately Unit 2 again — my ADHD therapeutic toy — and worse, it answered the wrong question. A face invites you to ask how the object feels about you. The whole pivot in RE-I1 was away from that. I had drawn the thing I had just spent two months arguing against.

That is when the egg arrived, and it arrived as a negative decision rather than a positive one. I was not looking for the friendliest shape. I was looking for a shape that could not become a dashboard and could not become a pet. An egg has no front. It has no face to read, so you cannot make eye contact with it, so you have to read the whole body — which is exactly how the field participants described noticing their animals: not from a signal in one place, but from the overall state of the thing.
Two more shapes died on the way. A bare board on a stand, which is honest but has no presence in a room. And a screen embedded in the surface, which is a dashboard wearing a costume.

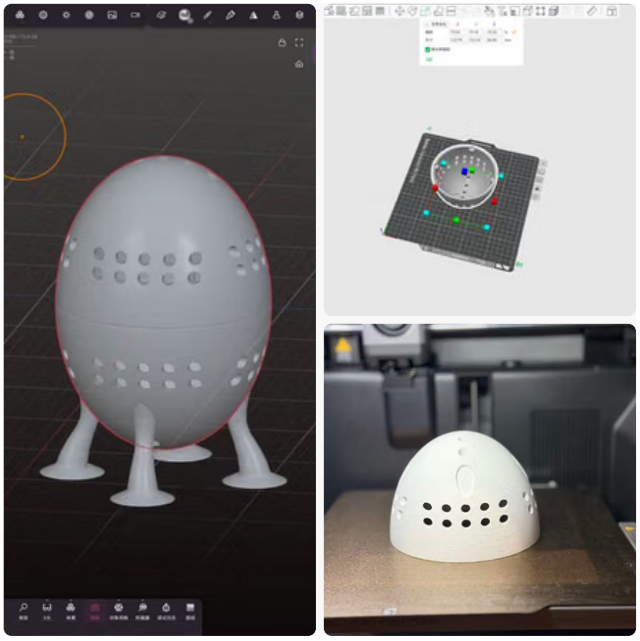
> *The cost I am not hiding: this is a closed body. Two halves meeting as one form, held by magnets, no vent, no fan, no opening. Inside it a Raspberry Pi runs the emotion network continuously and a battery module feeds it, and a closed curved shell is close to the ideal shape for keeping heat in. I have measured that the Pi stays within safe range under load, and I still printed it without any thermal provision at all. RE-I16 is where I record why, and that I have not paid that bill yet.*
> *The shell is perforated — both halves carry a field of round holes — but those are apertures for light, not vents. Where they sit and how big they are was decided by how the light should look, not by where the heat is. A hole that*

> *happens to let some air through is not a cooling design, and calling it one afterwards would be the kind of retrofitted justification this Padlet is meant to catch.*

TYPE 🟨 **INSIGHT**
DATE week 1
ID RE-I15
Status LOAD-BEARING

0

Modelling the Shell — Turning an Argument Into a Printable Body



RE-I15 decided the egg as an argument. This card is where the argument had to become a mesh — a 3D file made of triangles that a printer can actually understand — and where I found out which parts of the argument were expensive.

The brief I gave myself was short. It has to read as one continuous body rather than an assembly of parts, because RE-I15 says the whole point is that you read the entire object rather than a face or a screen. It has to open, because everything inside it is a prototype and prototypes get rewired. It has to let probes out into soil or water without the opening becoming the visual centre of the object. And it has to hold a Raspberry Pi, an Arduino, a battery tower and a ring of thirty LEDs, which between them are much less compact than the drawing suggested.

The first real conflict showed up immediately: a continuous surface and a serviceable interior want opposite things. Every seam I add to make it openable is a line across the body that the eye follows, and the eye following a seam is the eye reading the object as a machine again. Every seam I remove makes the inside harder to reach.
What I did not do is resolve that tension in the model. I chose to let the seam exist and to place it where the object is least likely to be looked at, and I am recording that as an unresolved compromise rather than a solved problem. RE-I15 already committed me to naming what this shape costs.

TYPE 🟩 MAKING
DATE:week7
ID RE-M5
Status SUPPORTING

0

I Knew It Would Get Hot and I Printed It Anyway — Three Days Before the Deadline

This card is about a decision I made under time pressure and would defend again, and about the specific thing it cost.

RE-I15 chose a closed body over a vented one, and I still think that is right — a vented shell reads as equipment, and equipment is not what this object is trying to be. So this is a sealed form: two halves that meet as one continuous egg, held shut by magnets, no vent, no fan, no opening. Inside it a Raspberry Pi 4B runs the emotion network continuously, fed by a battery module delivering up to 4 amps, and a closed curved shell is close to the ideal shape for keeping heat in. I knew that while I was modelling it.

I did not design anything into the shell to deal with it. No heatsink cavity, no thermal path to the outer surface, no mounting for a metal block, no channel for air to move.

One thing I want to be precise about, because a photograph will raise it. The shell is covered in holes. It would be easy to look at it and assume that is ventilation, and easy for me to let that assumption stand.

It is not. Those holes are where the light comes out. Their positions and sizes were decided by the pattern I wanted visible from across the room, and they sit in the upper dome and around the waist — nowhere near the processor, which is low in the lower half, underneath the battery and the wiring. There is no path from the hot part of this object to any opening in it. Air is not moving through it in any direction I designed or can predict. So the accurate description is a closed body with decorative perforations, and the perforations do not help. If some incidental air exchange happens through them, that is luck, and luck is not a thermal strategy.

The reason I designed nothing is not that I overlooked it. It is that I was three days from the portfolio submission and had not started laying it out. A print of the full body takes hours. A revised body means a revised print, a revised fit check, and a revised assembly, and if it had gone wrong I would have had no object to photograph and no photographs to lay out. So I chose to print the shape I had and spend the remaining time on the submission.

That is a real trade and I want it written as one rather than dressed up. I traded the object's ability to run for long periods against having an object at all on the day it had to exist. Given the same three days I would make the same call, and it is worth noticing that this is precisely the kind of pressure that makes people quietly stop mentioning a known problem — which is exactly why it gets its own card instead of a footnote.

What I have measured: the Pi stays within its safe range under load. That test was on a bench, over hours, with the shell open. "Safe on a bench for hours" and "safe sealed inside a curved shell for a week" are different claims and I have only earned the first. Every demonstration in this Padlet runs for minutes, so nothing I show is affected — but every long-deployment ambition in RE-Q4 and RE-M26 is, and those are the ones that matter for what this object is actually for.

What the next revision does, already worked out and not yet printed. A passive heatsink — a finned metal block sitting on the processor, pulling heat away by conduction — with a mounting boss modelled into the shell so it sits properly rather than being wedged in. Passive specifically, not a fan, and the reason is not thermal. A fan is audible. An object whose entire argument is that its state should be read from the outside cannot have a whirring noise that comes on when it is working hard, because then the noise becomes the primary signal and the light becomes decoration. It would also be genuinely absurd for a device with a microphone in it to be the loudest thing in the room.

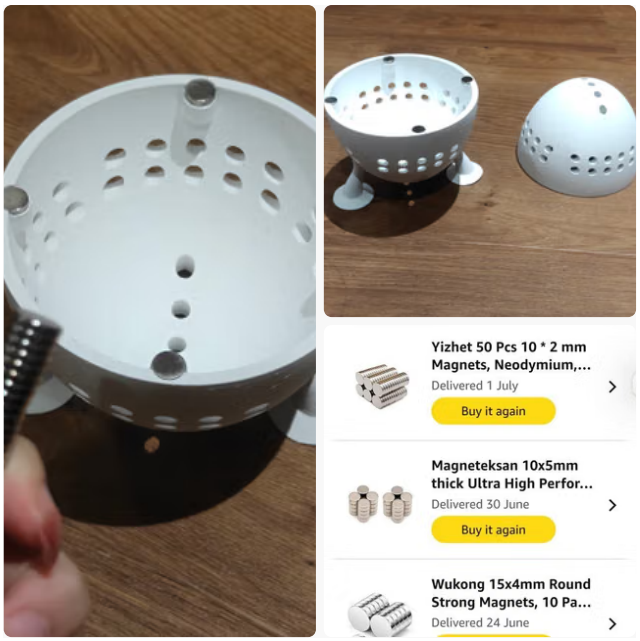
So: a known problem, a deliberate deferral for a stated reason, a solution designed and not yet built — and a shell full of holes that look like they solve it and do not. Marked RUNNING rather than SUPPORTING, because RE-S1 defines RUNNING as unfinished or paused on purpose, and this is exactly that.

TYPE 🟨 INSIGHT

DATE 7.5
ID RE-I16
Status RUNNING

0

Three Reprints, Three Sizes of Magnet



The body is two halves that meet as one continuous egg. No hinge, no catch, no screws, no visible fastener. You lift the top off and put it back, and closed it reads as a single form rather than as a container with a lid.

That is held by magnets set into the rim of both halves, and getting there cost me three complete reprints and three separate orders of magnets.

Here is how that happened, because the shape of the mistake is the point.

I printed the body. I placed the magnets. Then the dimensions changed — the interior had to grow to fit what was actually going inside, which is a Raspberry Pi, a printed case for the Arduino, a battery module, and a great deal of jumper wire that takes up far more room in reality than in a model. So I reprinted at a new size. The rim is now a different thickness, which means the recess for the magnet is a different depth, which means the magnets I already own do not sit flush. Order a new size. Print again. Same thing happens.

By the third one I had three different sizes of magnet in three little bags, and the working set is the third.

What I want visible in this is that the failure was not in the magnets. It was that I kept treating the closure as a detail to be resolved after the body, when in fact the rim thickness, the magnet depth and the interior volume are one problem with three names. Each time I changed one I quietly invalidated the other two, and each time it cost a print run of several hours to find out. This is the same error as RE-M8 — sizing an opening from the model rather than from the physical part — arriving in a different disguise.

Why magnets at all, having paid for them three times.

I open this thing constantly. Any closure that punishes being opened will eventually make me stop opening it and start guessing, and guessing is how a project starts describing an object that is no longer the object on the desk. Screws mean finding a screwdriver at midnight. A printed snap fit fatigues and eventually cracks, and RE-M8 is already a card about how a badly fitting printed part becomes an intermittent fault I will later blame on my code.

And magnets do one thing the other two cannot: they let it be opened in front of somebody, in one motion, without tools. That matters more than it sounds for a project whose whole position is that its process should stay visible. An object you can lift the top off in front of a person is an object whose insides are part of what it is, rather than something sealed behind fasteners.

The lower half stands on three conical legs. It does not sit on a surface, it stands above one — which keeps the probe cable exit off the table and, more importantly,

stops it reading as a device that has been placed down and starts it reading as something that is standing there.

TYPE 🟩 MAKING

DATE 7.5
ID RE-M6
Status DONE

0

Light Through Holes — Not a Diffuser, a Perforated Skin

Inside the upper half there is a strip of individually controllable LEDs running around the inner rim. Bare, that is a row of bright dots, and a row of bright dots reads as an appliance. A router has lights. A charger has lights. Nothing in a room full of indicator lights invites you to read it as a state.

Closed, the body reads as one continuous form — two halves meeting as a single egg, which is what RE-I15 asked for. What it does not do is glow as one continuous surface. The light leaves through a field of round holes drilled through both halves, so what you see from across the room is a single object carrying a pattern of lit points, not a smooth luminous shell.

The obvious alternative was a diffuser — a translucent layer that scatters the light so the surface glows and the sources disappear. I did not build one.

Partly that is time. Perforating a printed body was something I could do in the days I had; a properly diffusing double-wall shell was not. But having lived with it I would argue for it now, and here is the argument.

A perfectly even glow has no body. It is a lamp, and it is what every smart speaker and every air purifier already looks like — which is the exact reading RE-I15 was trying to escape. A perforated shell has depth. The holes near the strip are bright, the ones further round are dim, and as the brightness breathes the whole pattern moves unevenly rather than pulsing as one flat plane. It looks like something lit from inside rather than something that is a light.

Two things I want to keep straight, because both are easy to get wrong from a photograph.

These holes are apertures, not vents. Where they sit and how large they are was decided by how the light should look, and there is no path from them to the processor. RE-I16 is the card about heat, and it says plainly that the perforations do not help with it.

And a pattern of points is a different kind of reading from a surface. I designed the mapping, so I cannot tell whether a stranger sees one body changing state or a set of dots that happen to move together. That is worth asking someone rather than deciding myself, and it

belongs to RE-P4. The question is not "do you like it." It is "did you read it as one thing."

TYPE 🟩 MAKING

DATE 7.5
ID RE-M7
Status DONE

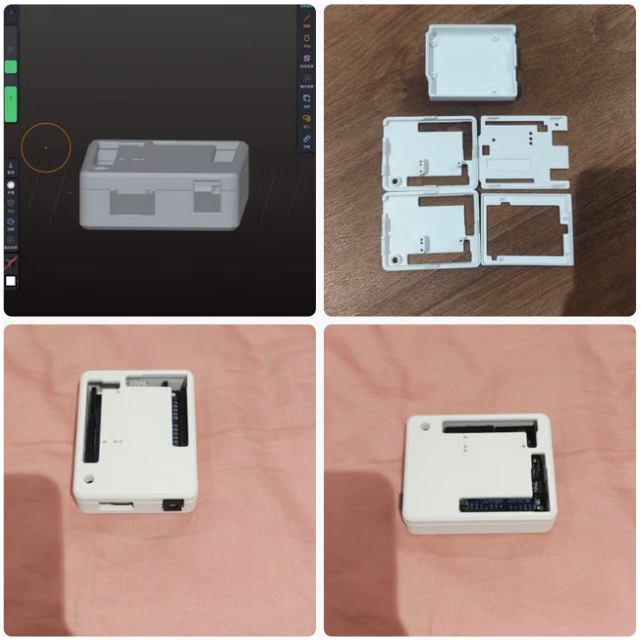
0

0

mmexport1782951176363.mp4

Five Failed Shells for One Plug, and Why I Kept Printing：arduino uno r3 usb c case

The Arduino Uno board has one USB-C socket on it, and that socket has to poke out of the egg — it is both how the board gets power and how it talks to the Raspberry Pi inside. So I needed a small printed collar around that socket: strong enough to hold the plug straight, thin enough to fit the curve of the shell. I printed it. It did not fit. I adjusted and printed again. Did not fit. Third, fourth, fifth. Each one took most of an hour on the printer, and each one was wrong in a different way — too tight to seat the plug, too loose to hold it, wall too thick to sit flush against the curve, opening in slightly the wrong place. The sixth one fitted.
Three shortcuts were available the whole time and I want them on the record, because refusing them is the actual decision here.
I could have left the sensor board hanging outside the shell entirely. Instant success photograph, and the object in that photograph would not be the object I have been describing for eight cards.
I could have run only a power cable in and left the data out of it. That would have quietly broken the architecture in RE-I9: the two boards do not just need electricity, they need to talk. A power-only connection turns the egg into a thing that glows, not a thing that translates.

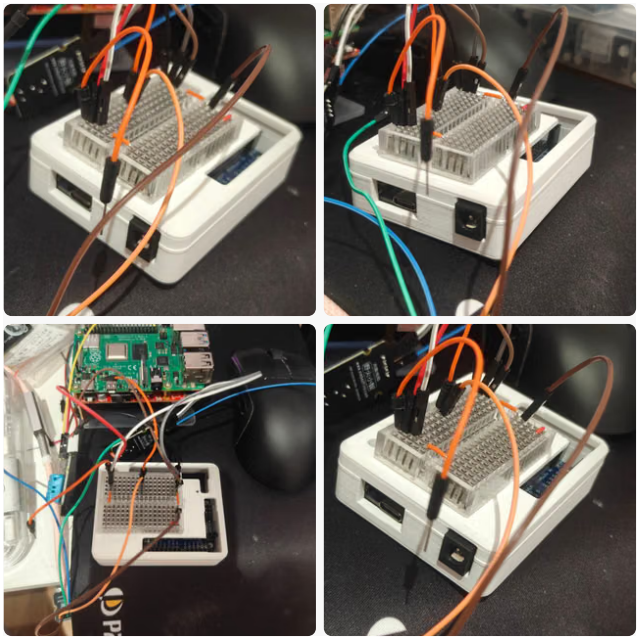


I could have glued the fifth one in place. It would have held for the presentation and then, six weeks later, the light would start flickering and I would spend three days blaming my code for a mechanical fault. I kept the five dead shells. They are in the documentation photographs on purpose. A pile of failed plastic is more informative about how I work than one clean render, and this project has been arguing since Investigate that the seams should stay visible.

TYPE 🟩 MAKING
DATE week7
ID RE-M8
Status DONE

0

0



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generated by chatgpt5.5

0

Probe Design — The Part That Has to Leave the Body

Everything else in this object is protected. The probes are not, and that changes how they have to be designed.

What actually goes out. The capacitive soil moisture probe — capacitive meaning it senses water through an insulated surface rather than by passing current between two exposed metal prongs, which is why it does not corrode into nonsense after a few weeks. The conductivity probe, which is waterproof and lives on a two-pin lead. And, when it is finally wired, the pH probe, which is a glass bulb on a BNC connector — a screw-lock plug — and is by some distance the most fragile object in the whole build.

What that forces. The probes cannot be permanent, because a glass pH bulb has a shelf life and a moisture probe can simply die — mine did, on the first one I bought, which is written up in RE-I30. So the exits have to be openings the probe passes through and can be pulled back out of, not holes the probe is glued into.

TYPE 🟩 MAKING
DATE 2026.7.5
ID RE-M10
Status RUNNING

0

HARDWARE

Hardware Preparation — Parts Land on the Desk

What I was deciding:
Who does the feeling math, and
who does the touching of the world?

Options I put side by side
- One computer for everything
  Smart enough for the “feeling” part, but clumsy at reading many cheap sensors at once. I would spend the whole month inventing adapters instead of making a thing.
- One tiny board for everything
  Great at sensors. Too small to run the feeling system I already built in Investigate. I would have to throw that work away or hide it in the cloud.
- A phone-like board that needs the internet
  Looks modern. Dies when Wi-Fi dies. The object stops being a presence in the room and becomes a guest of the router.
- Two boxes: a small board that only senses + a bigger board that only feels ← chosen

Why that leap:
Investigate already gave me a working “feeling engine.” Realise’s job is not to reinvent it. Realise’s job is to give it hands. Splitting the work lets each half stay honest.
What this shows about my thinking:
I am willing to make the build look less “one elegant chip” if it protects the research story. Elegance that erases Investigate is fake progress.

TYPE 🟩 MAKING
DATE week 7
ID RE-M1
Status SUPPORTING

0

Two Boards, Not One — Splitting the Nerves From the Brain

The first hardware question was embarrassingly basic: what actually goes inside.

My instinct was one board doing everything, because one board is elegant and elegant is easier to photograph. I looked at three versions of that and none of them survived.

One Raspberry Pi doing everything. The Pi (a small full computer, about the size of a credit card) is strong enough to run the CfC emotion network I built in Investigate. But it is bad at reading a lot of cheap analogue sensors — it has no analogue inputs at all, so every single sensor would need its own converter chip. I would have spent the whole Realise window building adapters instead of building a thing.

One Arduino doing everything. The Arduino (a much simpler board, roughly a very small brain that does one job forever) is excellent at sensors — that is what it exists for. It is nowhere near powerful enough to run a neural network. Choosing it means throwing away Investigate's core, or hiding it on a server somewhere, which is the same as throwing it away.

A board that streams everything to the cloud. Modern-looking, and dead the moment the Wi-Fi drops. The object stops being a presence in the room and becomes a guest of the router. I had already watched the wild-type experiment lose continuity to a crash-and-restart seam; I was not going to design a new dependency that breaks the same way.

So: two boards. The Arduino Uno is the nervous system — it touches the world, reads the sensors, and does nothing else. The Raspberry Pi 4B is the brain — it does the feeling and the light, and never touches a sensor directly. Between them, one USB cable carrying one line of text every two seconds.

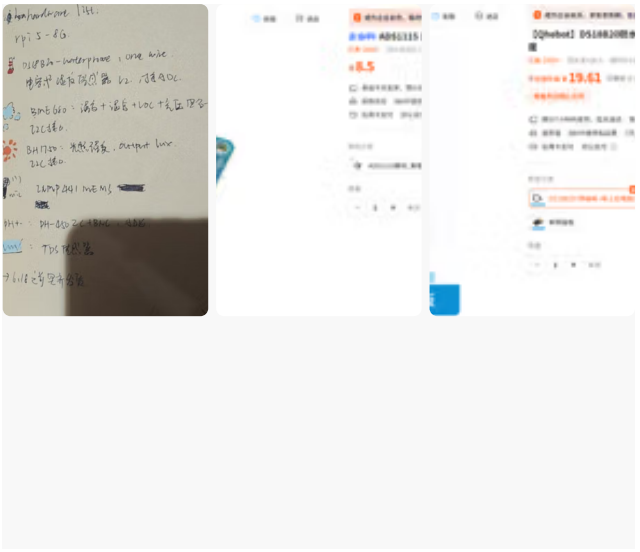
The leap here is not technical, it is about what I was willing to protect. Investigate already produced a working feeling engine. Realise's job is not to reinvent it — Realise's job is to give it hands. The moment I framed it that way, the "one elegant chip" version stopped being attractive, because elegance that erases Investigate is not progress, it is amnesia.

TYPE 🟩 MAKING

DATE week7
ID RE-M1
Status SUPPORTING

0

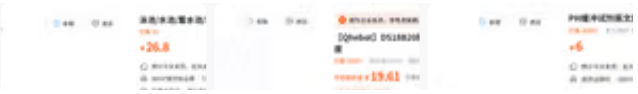
The Sensor Wish-List Versus What Is Actually Plugged In



This card exists so the shopping list and the desk never get confused with each other. For each sense the project claims, here is what I wanted, what is physically wired right now, and what I cut.

Temperature. Wanted a DS18B20 probe (a sealed metal temperature probe you can push into soil or water). What is wired is a DHT11 (a cheap combined temperature-and-humidity module) plus a BMP280 (a pressure-and-temperature chip). They work, but the DHT11 reads in whole degrees with about two degrees of error — coarse for a project whose whole idea is noticing small

departures from an environment's own rhythm. Written down as a known weakness, not smoothed over.



Soil moisture. Wanted a capacitive probe — one that senses water through an insulated surface rather than by passing current between two exposed metal prongs. The cheap pronged kind corrodes and then reports nonsense, so I refused those early. I have the capacitive one, it is wired to pin A1, and it is the one sensor in this project that is genuinely calibrated: on 5 July I measured it in open air (raw value 0–4) and submerged in water (350–370) and put those two numbers into the firmware. Worth saying that the first one I bought was simply faulty. I spent a day suspecting my wiring and then my code before testing the part itself and finding it dead. Replacing it fixed everything. That day is the reason I now test the component before I debug the system.

Air and smell. Wanted a BME680 — one chip that reports temperature, humidity, pressure and VOC (volatile organic compounds, roughly "the chemical staleness of air"). What I have is the BMP280, which does not sense VOC at all. So there is currently no smell sense on this device. The field is simply absent from the data stream rather than filled with a plausible-looking zero.

Light. Wanted a BH1750 (a digital light sensor that outputs real lux values straight away). What is wired is a photoresistor — a component whose resistance changes with brightness, giving a raw number I have to convert myself with an approximate formula. Fine for showing a day-night rhythm. Not something I would put in a table of measurements.

Sound. Wanted an INMP441 (a digital microphone). The Arduino Uno cannot receive that kind of digital audio, so I switched to a MAX4466 — still a real microphone, just one that outputs a varying voltage instead of digital audio. This one is settled, not a compromise I regret. There is a task left on it: a single instantaneous reading of a microphone is meaningless, so the firmware has to sample continuously for a few tens of milliseconds and take the difference between the loudest and quietest point. That is written in my to-do list, not done.

Water chemistry. The pH probe (PH-4502C) is bought and still not wired — the pin is reserved and the firmware flag is switched off, so the field is omitted entirely rather than sent as zero. Sending zero would be read as extreme acid, which is worse than saying nothing. The TDS/EC probe (conductivity, which stands in for how much dissolved salt or fertiliser is in water) is wired and running on the manufacturer's default formula, with no calibration against known reference solutions. Marked RUNNING, not DONE.

Human presence. There was a PIR motion sensor in the early sketches, from when this was still a companion object. Removed. It answers "has someone come near" — a question about you, not about the environment. It was the last piece of the pet brief still physically attached to the board, and taking it off was the cheapest and most decisive thing I did all week.

TYPE 🟨 INSIGHT

DATE 2026.7.5
ID RE-I30
Status LOAD-BEARING

0

Power — Can This Thing Sit Next to Soil, or Only Next to My Laptop?

This looked like a boring question until I noticed it decides whether the object is finished or not.

A wall charger works perfectly and ruins the claim. If the egg has a cable running to the nearest socket, it is a desk prototype, and a desk prototype cannot be placed next to a plant, a tank, or in the middle of a room, which is where the whole argument about ambient reading lives.

I tried a phone power bank next. It failed in a specific and annoying way: power banks are designed to cut out when the device stops drawing much current, so it kept switching itself off during quiet moments — exactly the moments the object is supposed to be sitting there doing very little. A power source that punishes stillness is the wrong power source for a thing whose main behaviour is stillness.

What I am using now is a 21700 battery tower module (a stack of large rechargeable cells with a board on top that outputs a steady 5V at up to 4A), connected to the Pi through pogo pins — spring-loaded contacts that press against a pad rather than plugging in, so there is no connector to snap off inside the shell. It has been powered up and verified. It matches what the Pi actually draws, and it is one object rather than a knot of cables, which matters when the whole thing has to fit inside a sealed egg.

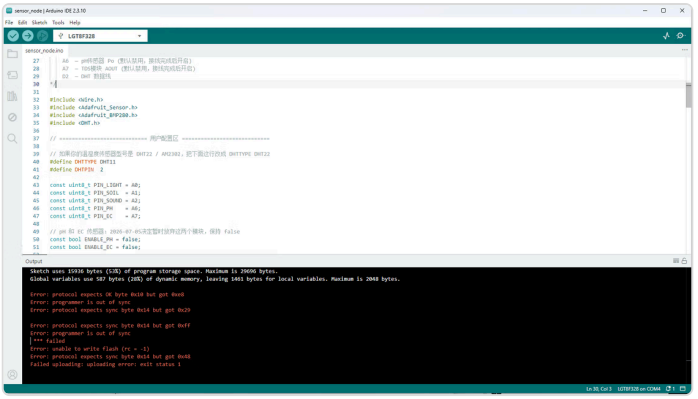
The honest limit: batteries run out. When I write up a demonstration I have to state how long it ran on its own. "Cordless" in this project means allowed to live in the room, not immortal — and if I ever show a photograph of it sitting in soil with no cable, the caption has to say for how long.

TYPE 🟩 MAKING

DATE week7
ID RE-M9
Status SUPPORTING

0

The Board Was Not What the Label Said



I bought what was sold as an Arduino Uno R3 — the standard beginner board, the one every tutorial on the internet is written for. I wired the sensors onto it, wrote the firmware, and tried to load the program onto it from the Raspberry Pi, so that the whole egg could be self-contained.

The first handshake worked. The board answered. Then the actual writing of the program failed, every single time, at exactly the same stage. I did the obvious things — different cable, different port, different speed setting — and it failed identically each time, which is the specific kind of failure that means you are wrong about something more basic than what you are testing.

When I gave up and plugged it into a Windows laptop instead, the Arduino software identified the chip as an LGT8F328, not the ATmega328P that a real Uno carries. It is a compatible replacement chip made by a different manufacturer: same pin layout, mostly the same behaviour, different silicon underneath, and — importantly — a slightly different protocol for accepting a new program. Combined with the USB-to-serial chip on this particular clone (a CH340, whose Linux driver is known to be awkward with large data transfers), that explained the failure completely.

Three ways I could have gone.

Keep insisting it is a real Uno and keep debugging against a false model. That is how you lose a week.

Throw it away and buy a genuine board. Cleaner, and it spends money and days I do not have, and it discards everything I just learned about how this specific board behaves.

Accept what it actually is, load programs onto it from the Windows laptop, and let the Raspberry Pi only read from it. That last part is what I did, and it is not a defeat — the Pi never needed to write to the Arduino in normal operation anyway. It only needed to listen. So the workflow is slightly more awkward at build time and identical at run time.

What this changed in how I work: a label on a product page is a claim, not evidence. I now check what the hardware says about itself before I trust what the seller says about it. I would rather write an ugly true sentence than a tidy false one.

TYPE 🟨 INSIGHT

DATE week7
ID RE-I18
Status LOAD-BEARING

0

It Still Failed on the Laptop — I Blamed the Cable, the Answer Was a Dropdown

Read this one directly after RE-I18, because the whole point is that I thought I had already solved it.

I had just worked out that the board is an LGT8F328 clone and that I should load programs onto it from the Windows laptop rather than the Raspberry Pi. So I plugged it into the laptop expecting relief. It compiled fine and then failed while writing to the chip, throwing an error that reads "not in sync" and spitting back meaningless bytes.

Here is my detective board, in the order I actually worked through it.

The cable. Obvious first suspect, cheap to test, and cables really do fail this way. Swapped it. Same failure.

The port. Unplugged, replugged, tried every socket. Same failure.

Power. Wondered whether the board was browning out mid-write. Tried a powered connection. Same failure.

Speed. Lowered the transfer speed, because "it works until it has to move a lot of data" smells like a timing problem. This felt like the sophisticated answer. It mostly still failed.

Then the actual cause, which was in a menu. Arduino's software has a setting called Clock Source — which crystal the chip should assume it is running from, effectively how fast the board's internal heartbeat is. The software's default for this chip family is Internal 32MHz. This particular board runs from an external 16MHz crystal — the product listing said so, in a line I had read and not understood. So the software was talking to the board at one tempo while the board was keeping time at another. Every message arrived slightly out of step, which is exactly why the handshake succeeded (short, forgiving) and the bulk write failed (long, unforgiving).

I changed one dropdown from 32MHz to 16MHz. It worked immediately, and has worked every time since. Confirmed 5 July.

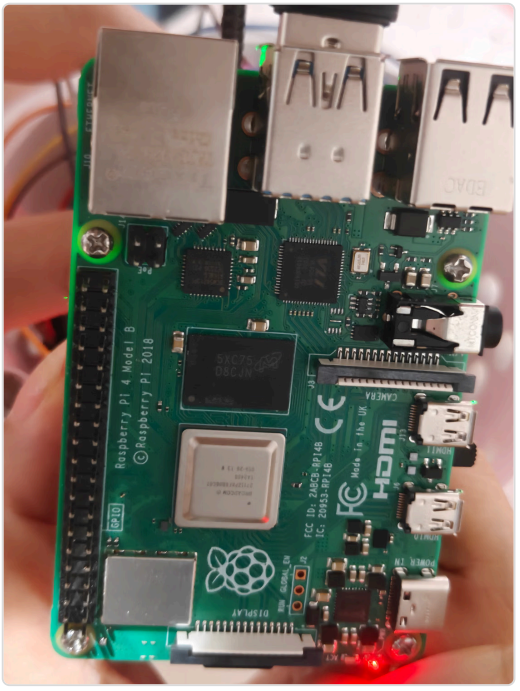
What I want visible here is not the fix, it is the shape of the mistake. I spent my time on the glamorous physical suspects — bad cable, bad power, bad port — because those are the failures that feel like real engineering. The actual fault was ordinary, invisible, and sitting in a menu I had never opened. The useful skill was not knowing about clock sources. It was noticing that the failure was too identical to be a physical fault, and being willing to kill my favourite theory.

TYPE 🟨 INSIGHT

DATE 2026.7.5
ID RE-I17
Status LOAD-BEARING

0

Two Identical Sockets, One of Them Cannot Talk



This is the most ordinary mistake in the whole build and probably the easiest one for anyone to understand, which is why I want it written down properly rather than buried.

The Raspberry Pi has USB-C sockets. They look the same. One of them has the word POWER printed next to it in small letters. That one is wired only for electricity — it can feed the board, and it physically cannot carry data. The others carry both.

I plugged the Arduino into the POWER one.

The result is the cruellest possible failure mode: everything looks alive. The Arduino's light comes on. The board is clearly getting power. Nothing is broken, nothing smells hot, no error appears. There is simply no data, ever, and the software just sits there waiting for a line that will never arrive. If you do not already know this distinction exists, there is nothing on the screen or on the desk to tell you.

I lost time to it, then found the
printed word, then moved one cable.

What I took from it is not "read the labels." It is that a lot of debugging time goes into problems that are invisible by construction — where the system's outward appearance is identical in the working and the broken case. That is exactly the situation this whole product is trying to fix for a different subject. A pot of soil that is drying out looks identical to a pot of soil that is fine, right up until it does not. ANIMA exists because some states have no surface. It is a decent joke on me that I spent an afternoon inside one.

TYPE 🟨 INSIGHT

DATE 2026.7.2
ID RE-I19
Status LOAD-BEARING

0

Writing the Contract Before Touching a Wire

Before any sensor went onto a pin, I use AI to wrote a document called SERIAL_PROTOCOL.md and made it the only thing the hardware side and the software side had to agree on.

The contract is short. Every two seconds, the Arduino prints one line of JSON — JSON is just text formatted so a program can read it, and one line looks like this: temperature 22.3, moisture 55.0, light 15200, and so on. It goes over the USB cable at 115200 baud, which is simply the agreed speed both ends run at. Field names are fixed. Order does not matter.

Two clauses in that document are doing real work.

First: the Arduino must send physical units, not raw numbers. A sensor natively produces a number between 0 and 1023 which means nothing on its own. The conversion into actual degrees or actual percent happens on the Arduino, before it is sent. So the software side never has to know what kind of sensor is on the other end — it receives meaning, not voltage.

Second, and this is the one I care about: if a sensor is not connected, the Arduino does not send that field at all. It does not send zero. It does not send null. The field is simply absent.

That clause exists because of pH. On the pH scale, 7 is neutral and 0 is battery acid. If an unconnected pH sensor reported 0, the software would faithfully translate that into "this water is catastrophically acidic" and the light would go red, and it would be right to, given what it was told. A missing sensor pretending to be a reading is not a small bug — it is the machine confidently describing an emergency that does not exist.

```
# 串口数据契约（Arduino → RPi）
## 这是硬件端和软件端唯一需要提前对齐的接口，请在开始接线/编代码前先看这份文件

---

## 1. 数据格式：每行一个 JSON 对象，通过 USB 串口发送

Arduino 每隔固定周期（建议 2 秒）向串口打印**一行 JSON**，以换行符 `\n` 结尾：

```
{"temperature":22.3,"moisture":55.0,"voc":12.4,"humidity_air":48.0,"light":15200,
```

字段名固定，顺序不重要，缺失的字段直接不写这个key（不要写null/None）。

字段名	对应传感器	单位	类型
`temperature`	DS18B20	°C	float
`moisture`	电容式土壤湿度	百分比(0-100，或原始ADC值，见下方"标定"说明)	float
`voc`	BME680	ppm(近似)	float
`humidity_air`	BME680	百分比RH	float
`light`	BH1750	lux	float
`sound`	声音传感器	相对强度(0-100)	float
`ph`	PH-4502C	pH值	float
`ec`	TDS模块	ppm	float

2. 波特率

Python 两端必须一致。Arduino代码里 `Serial.begin(115200);`, Pytho
```

Padlet Drive

SERIAL_PROTOCOL

So the rule is: absence stays absence. The software treats a missing field as "cannot see this right now" and feeds a neutral value, and it writes a warning to the log rather than silently smoothing over it. That single decision is the reason RE-I30 can honestly say there is no smell sense on this device instead of quietly outputting a plausible number.

Writing the contract first also meant that when the wiring took three weeks longer than planned, the software half was never blocked — it ran against simulated data in exactly the format the real thing would eventually send.

TYPE 🟩 MAKING

DATE 2026.7.1
ID RE-M12
Status LOAD-BEARING

0

I Worked the Pins Out in Conversation with an AI

There is no wiring diagram in this project. I want to say that first, because "AI-assisted wiring" sounds like a picture was generated and it was not.

What actually happened is less visual and more ordinary. I had a list of pins on the board and a pile of physical modules on the desk, and I worked out the assignment by talking it through with an AI assistant — which sensor goes on which pin, what voltage each one wants, which ones can share, which pins were already taken. The output was a decision, written into my own notes, not an image.

I am disclosing that the same way AL-P0 discloses how the field interview instrument was built. A tool that shapes the work should be visible in the account of the work, and this one shaped several hours of it.

What it was genuinely good for. It knew that two of my sensors both wanted the same pair of pins and could legitimately share them, because that pair is a shared two-wire connection where each chip answers to a different address — that is the only reason everything fits on a board this small. It knew that the soil moisture module runs on 3.3 volts and not the 5 volts most of the others use, which is the kind of detail that quietly destroys a component. And it was faster at holding the whole pin list in view at once than I was, which is exactly what I wanted from it.
What it got wrong, and what I therefore had to verify myself.
It reasons from the part name I give it. I named the parts from my ideal specification rather than from my desk, so it happily assigned pins for a BME680 and a BH1750. I have a BMP280 and a photoresistor. Everything downstream of those two names had to be redone by hand against the modules I actually own, and RE-I30 is the card where I finally sat down and wrote out the difference between the shopping list and the desk.

It did not know the board was an LGT8F328P clone rather than a genuine Arduino Uno, because I did not know that yet either — see RE-I18. Nothing in the pin assignment was wrong because of it. But every statement it made about that board carried a confidence that turned out to be unearned, and I could not have detected that from the conversation. And it could not tell me which of the Raspberry Pi's USB-C sockets carries data. That is not a reasoning problem. It is a physical fact about one specific board sitting in front of me, printed in small letters on the board itself, and it cost me an afternoon — RE-I19.

The pattern underneath all three is the same, and it is the reason this card is worth writing. The AI has never seen my desk. Everything it says is true of the part as specified, and my build is made of the particular ones I own — a clone chip wearing a famous name, a cheaper sensor standing in for the one in the document, a socket that looks identical to the one next to it. This project keeps running into the same wall from different directions: the specification is not the thing.
So the working rule I ended up with. The tool is fine. The disclosure is mandatory. The verification is mine. Every pin in the final build was confirmed against the physical module in my hand and written into the pin map in RE-S5, and that pin map — not any conversation — is the record.

TYPE 🟩 MAKING
DATE 2026.7.2
ID RE-M11
Status SUPPORTING

0

Wiring Light (A0) — A Sensor That Only Knows Brighter and Darker

The light sensor is a photoresistor on pin A0 — a small component whose electrical resistance falls as more light hits it. Wire it in a divider circuit and you get a number that goes up when the room gets brighter.

That is genuinely all it knows. It does not report lux, the actual unit of illumination. It reports a raw number that I have to convert with an approximate formula I wrote myself, and that formula is not calibrated against anything.

What I wanted was a BH1750, a digital light sensor that outputs real lux values directly, no conversion needed, no guessing. It is on the list and not bought.

Which leaves an honest split I want stated rather than fudged. Relative change over time — it is getting darker, it has been bright for six hours — is something this component reports reliably, and that is what the soil genome actually uses light for. Look at RE-M2: light is weighted 0.7 not because light is a threat to a root, but because a long bright stretch is what precedes the ground drying. For that job, a rough rising and falling curve is enough.

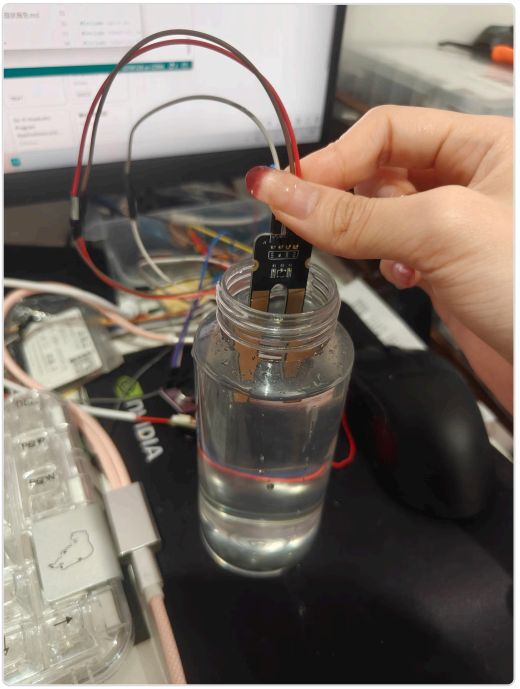
What I cannot do is put a lux figure in a table and call it a measurement. So the light channel gets used for rhythm and never for absolute claims, and this card is where that limit is written down so that no later demonstration card can quietly forget it.

TYPE 🟩 MAKING

DATE 2026.7.4
ID RE-M13
Status RUNNING

0

Wiring Moisture (A1) — The One Sensor That Was Dead



This is the sensor the whole soil demonstration rests on, and it is the only channel in the project that has been calibrated against physical reality rather than estimated from literature.
It is a capacitive soil moisture probe on pin A1 — capacitive meaning it senses water through a sealed surface rather than by passing current between two exposed metal prongs. The cheap pronged kind corrodes within weeks and then reports drift as if it were data, so I ruled those out before buying anything.
One detail that cost me time and is worth knowing: this probe runs on 3.3 volts, not 5. It has a Grove connector with four wires, one of which is unused, and connecting it to 5V the way most sensors on this board are connected will damage it.
Then the actual story of this card. I wired it, and the readings were nonsense — not noisy, not drifting, just meaningless. I assumed it was me. I checked the wiring. I checked the pin assignment. I rewrote the conversion in the firmware. I checked the voltage. I spent a day working through my own possible mistakes, in order, from most likely to least.
The sensor was simply faulty. A new one, wired identically, worked immediately.

I have kept that day in the account because of what it changed rather than what it cost. I now test a component in isolation before I debug the system it sits in, and that habit is exactly the reverse of what feels natural — it feels arrogant to suspect the part before suspecting yourself. It is also what a field participant described about noticing their animal: the skill is knowing which of the possible explanations to check first, and that ordering is learned expensively.
Calibration, done 5 July: in open air the probe reads a raw value of 0 to 4. Fully submerged in water, 350 to 370. Those two numbers went into the firmware as SOIL_DRY_RAW and SOIL_WET_RAW, and everything between them is interpolated. Worth noting this particular unit reads higher when wetter, which is the opposite direction from some other models — the conversion formula is written so the direction does not matter as long as the two endpoints are correct.

TYPE 🟩 MAKING
DATE 2026.7.5
ID RE-M14
Status DONE

0

Wiring Sound (A2) — Why One Reading of a Microphone Is Meaningless

The sound sensor is a MAX4466 on pin A2 — an electret microphone with an amplifier, which outputs a varying voltage rather than digital audio.

I had wanted an INMP441, a digital microphone. The Arduino Uno has no hardware support for the kind of digital audio connection that part needs, so I switched. I want to be clear that this is a settled substitution and not a compromise I regret: both parts answer the same question the project actually asks, which is roughly how much is going on around here.

The interesting part is what I got wrong about how to read it.

A microphone output is a wave. It swings above and below a middle resting value, thousands of times a second, and the loudness is how far it swings — not where it happens to be. So a single instantaneous reading of a microphone tells you nothing at all. It might catch the top of a wave during silence, or the exact midpoint during a shout. Reading it once and calling that "the sound level" is like judging how rough the sea is by measuring the water height at one random instant.

The correct approach, which is written into my firmware to-do list and not yet implemented: sample continuously for a few tens of milliseconds, keep the highest and lowest values seen in that window, and report the difference between them. That difference is the amplitude, and amplitude is loudness.

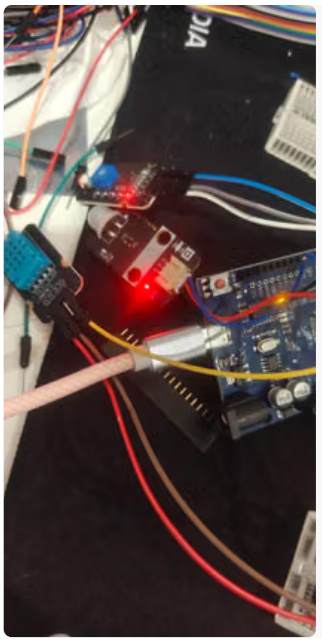
Marked RUNNING because it is a known task, not a finished one. It also matters less than it sounds for the demonstration I am actually planning, since sound is weighted 0.15 in the soil genome — a root is not troubled by a noisy room, and RE-M2 explains why I deliberately kept it low rather than letting the object start reacting to me walking past.

TYPE 🟩 MAKING

DATE 2026.7.4
ID RE-M15
Status RUNNING

0

BMP280 and DHT11 — Two Sensors Doing Half of One Sensor's Job



The specification asks for a BME680. That is a single chip that reports temperature, humidity, air pressure, and VOC — volatile organic compounds, which in plain terms is the chemical staleness of air, the thing you notice when you walk into a room that has been shut all day.

What is actually on the board is a BMP280 on the I2C connection (pins A4 and A5 — I2C is a two-wire arrangement that lets several chips share the same pair of wires) plus a DHT11 on pin D2.

Between them these two cover temperature and humidity, with overlap, and they cover VOC not at all. The BMP280 has no chemical sense of any kind.

Two consequences I have to live with and state.

The VOC channel does not exist. Under the contract in RE-M12, the Arduino simply omits that field, so the software knows it is blind there rather than being fed a comfortable zero. This is not cosmetic: VOC is weighted 0.9 in the indoor air genome, the single highest weight of any channel in any of my three genomes. It is written up properly in RE-M4, and it is the reason the indoor air genome cannot currently be demonstrated at all.

The DHT11 is coarse. It reports whole degrees with roughly two degrees of error. For a project whose entire premise, per RE-Q4, is noticing when an environment departs from its own recent rhythm, a sensor that cannot resolve anything finer than a degree is a real limitation. The freshwater genome weights temperature at 0.9 and treats warming as its main proxy for oxygen risk — resting that on a DHT11 is not something I would defend in a laboratory.

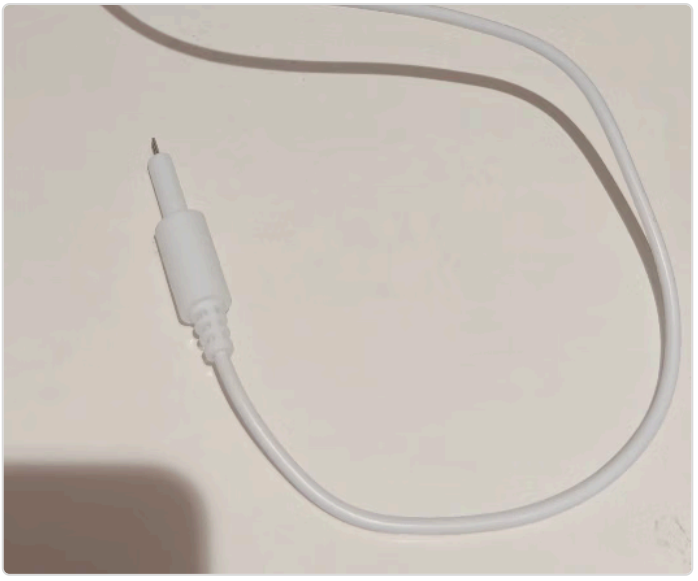
Both are marked as open decisions in the hardware report rather than quietly accepted, because buying a BME680 would fix both problems at once and the only reason it has not happened is money and time.

TYPE 🟩 MAKING

DATE 2026.7.4
ID RE-M16
Status RUNNING

0

Wiring Conductivity (A7) — Running on the Manufacturer's Guess



The conductivity probe is a TDS module on pin A7, wired on 5 July, with a waterproof probe on a two-pin lead. Conductivity means how easily electricity passes through the water, which rises with dissolved minerals — so it stands in for how much fertiliser or salt is in there. In the soil genome it separates two opposite problems that look similar from the outside: starvation, where there is nothing dissolved, and salt stress, where there is far too much.

It is wired, the firmware flag is switched on, and it produces numbers.

Those numbers are converted using the formula the manufacturer supplied in the product listing. I have not calibrated it. Proper calibration means buying standard reference solutions of known concentration — 342 and 707 parts per million are the usual pair — putting the probe in each, and adjusting the formula until the reading matches the known value. I have not bought them.

So this channel currently produces numbers that are probably in the right neighbourhood and are definitely not measurements. Marked RUNNING for that reason, and it is on the to-do list in the hardware report with the specific solutions named, so it is a defined task rather than a vague intention.

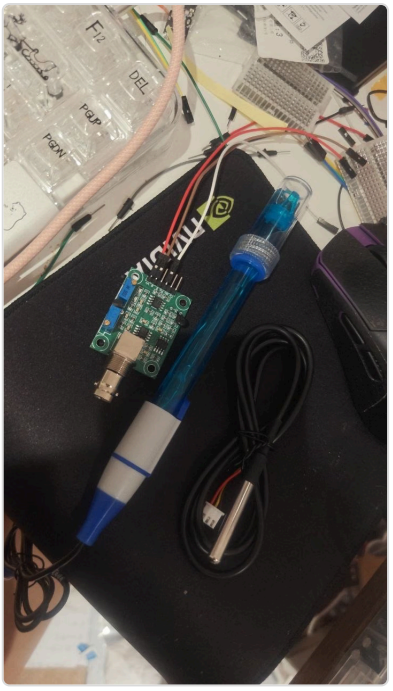
Why I wired it anyway rather than waiting: relative movement is still informative even when the absolute value is not trustworthy. If the reading climbs steadily over an hour, something is dissolving into that water, and that is true regardless of whether my conversion formula is off by fifteen percent. This is the same distinction I drew for light in RE-M13 — change is usable, absolutes are not — and it is worth noticing that I have now made that same call twice on two different channels, which suggests it is a principle rather than an excuse.

TYPE 🟩 MAKING

DATE 2026.7.5
ID RE-M17
Status RUNNING

0

pH Is Bought, Reserved, and Deliberately Not Connected



The pH probe — a PH-4502C board with a glass electrode on a screw-lock BNC connector — is sitting on my desk, not wired.

Pin A6 is reserved for it. The firmware has a switch called ENABLE_PH and it is set to false. Under the contract in RE-M12 that means the pH field is never sent, so the software knows it cannot see pH rather than being handed a number.

I want to be precise about why this is a card rather than an omission, because "I ran out of time" is true and is not the interesting part.

pH matters most to the freshwater genome, where it is weighted 0.8 — second only to temperature. So the genome that most needs this sensor is the one I am least able to demonstrate, which is stated plainly in RE-M3. That is not a coincidence. Water is the niche where the important variables are chemical, and chemical sensing is the expensive, fragile, calibration-hungry end of this whole project.

And there is a specific danger this card exists to prevent. On the pH scale, 7 is neutral and 0 is extreme acid. An unwired sensor that reported 0 would be read by the software as a catastrophe, the light would go red, and the demonstration would look dramatic and be entirely fictional. The contract rule about omitting absent fields was written with exactly this sensor in mind.

So: bought, reserved, switched off, and written down. The gap is visible in the account rather than papered over in the demonstration.

TYPE 🟩 MAKING

DATE 2026.7.5
ID RE-M18
Status RUNNING

0

Removing the Motion Sensor — Taking the Last Piece of the Pet Off the Board

There was a PIR motion sensor in the early parts list. PIR means passive infrared: it detects a warm body moving nearby. It is the component in every automatic light and every burglar alarm, and it costs almost nothing.

It was there from when this was still a companion object. The behaviour it enables is obvious and immediately appealing — you walk into the room, the egg notices you, it brightens. Everybody who has seen a prototype do that has smiled at it.

RE-I1 changed the brief and RE-Q2 sharpened the test: does the output report the environment, or audition for affection? A motion sensor cannot pass that test, because the only question it can answer is a question about you. Whether a person has come near is not a fact about the soil, the water, or the air. It is a fact about the person.

Two things make this worth its own card rather than a line in RE-I30.

The first is that it was the last piece of the abandoned brief still physically present. The pivot had been written up, argued, and cross-referenced back to AL-I19 — and this component was still

sitting in the parts pile the whole time, waiting to be wired. A design decision is not finished when you have written the paragraph. It is finished when the hardware agrees with the paragraph.

The second is that it was the cheapest possible decision to reverse and I did not reverse it. It costs almost nothing, the wiring is trivial, and it would have made every demonstration more charming. It is exactly the kind of small compromise that does not look like a compromise at the time. Every subsequent mapping decision in the software section gets tested against this one: if I would not accept a motion sensor, I cannot accept a rule that does the same thing in code.

Worth noting that this pressure did not go away when the component did. RE-I13 had to explicitly define the Connection dimension so that it responds to changes in the sensed medium and not to a person arriving — and my own portfolio deck still has the old wording in it, which is a live inconsistency I need to fix rather than a solved problem.

TYPE 🟨 INSIGHT

DATE 2026.7.4
ID RE-I20
Status LOAD-BEARING

0

The Firmware — Sample, Convert, Pack, Send, Wait

The program running on the Arduino is deliberately the least clever thing in this project. It has one job and no opinions.

Every two seconds it does five things. Read each connected sensor. Convert each raw number into a physical unit using that channel's calibration. Assemble the results into a single line of JSON. Print that line to the USB connection. Wait.

It does not decide anything. It does not know what a genome is, or what stress means, or that a neural network exists on the other end of the cable. It does not filter, average, or smooth. That is a design decision rather than laziness: every place where meaning gets added to a number is a place where I have made a judgement, and I want all of those judgements in one part of the system where they can be read, argued with, and changed by editing a text file. Splitting the interpretation across two boards means half of the reasoning is buried in firmware that has to be reflashed from a Windows laptop to change — see RE-I17 and RE-I18 for how enjoyable that is.

The parts of it that carry actual decisions, all of them recorded elsewhere: the moisture conversion uses the two calibration endpoints from RE-M14. Channels that are not wired are controlled by simple on-off switches — ENABLE_PH is false, ENABLE_EC is true — and switched-off channels are omitted from the JSON entirely rather than sent as zero, per RE-M12. The two-second interval matches what the software expects, and the speed is fixed at 115200.

One thing that is genuinely confusing and worth naming: this program communicates at 115200, but the process of loading the program onto the board happens at 57600. Two different speeds for two different purposes on the same cable. I mixed these up more than once early on, and the failure looks identical to a hardware fault, which is a theme by now.

Padlet Drive

sensor_node

TYPE 🟩 MAKING

DATE 2026.7.5
ID RE-M20
Status LOAD-BEARING

0

The First Real Packet

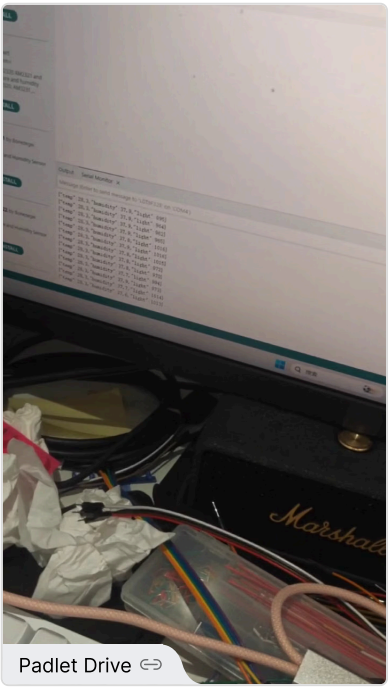
After the contract, the six printed shells, the clock source, the fake Arduino, the POWER socket, the dead moisture probe, and the calibration in a glass of water — the Raspberry Pi printed a line of JSON that had come from actual sensors reading an actual room.

That is the whole card, and I want it short on purpose, because the moment itself was short and slightly anticlimactic.

What it proves is narrow and I will not inflate it. It proves the chain is continuous: physical world, sensor, Arduino, cable, Raspberry Pi. It does not prove any reading is accurate. Most of them are not — the light value is an approximation, the conductivity uses a manufacturer's formula, the temperature is coarse, and the pH and VOC fields are absent.

What it changes is the status of everything downstream. Until this moment, every piece of software in the project had only ever been tested against simulated data — and I know exactly how much that matters, because the wild-type experiment in Investigate ran on a simulated world and had to be labelled as such in every card that touched it. The seam between "this works against numbers I generated" and "this works against a room" is the seam the whole Realise phase exists to cross.

The reading code has one feature worth mentioning: if the USB connection drops mid-run, it reconnects and carries on instead of crashing. That is there because the shell is closed and the cable is not strain-relieved yet, and an object that dies when it is nudged is not a presence in a room.



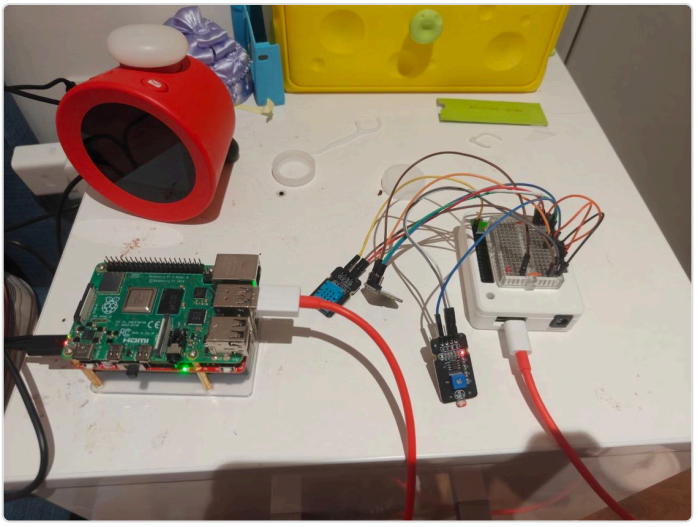
video_2026_07_02_00_54_13.mp4

TYPE 🟩 MAKING

DATE 2026.7.2
ID RE-M19
Status SUPPORTING

0

Setting Up the Raspberry Pi, and the Password I Have Not Changed



The Raspberry Pi 4B is the half of the object that does the feeling. Setting it up was the least eventful part of this build: write the operating system to a memory card, boot it, connect it to the network, and from then on work on it remotely from my laptop rather than plugging a monitor into a thing that is supposed to live inside a sealed egg.

Two things are worth recording.

The first is that remote access is not a convenience here, it is a requirement of the form. RE-I15 chose a closed body. A closed body means there is no screen, no keyboard, and no way in — so every future change to the software has to arrive over the network. The shape decision quietly determined the working method.

The second is a genuine fault that I am writing down rather than fixing silently. The Pi's password is currently 12345. That is on my home network, on a device that at the moment holds nothing but sensor logs, so the actual risk today is close to nothing. But this object is described throughout this Padlet as something that lives in a room, continuously, for a long time, listening through a microphone. A permanently networked microphone in a room with a five-digit password is not a small oversight in that framing — it is a contradiction between what I say the object is and how I have actually configured it.

It is on the to-do list in the hardware report. Writing it here rather than only there is deliberate: RE-I0 refuses to let the project look stronger than it is, and this is one of the places where the gap between the described object and the built object is embarrassing rather than technical.

TYPE 🟩 MAKING

DATE 2026.7.4
ID RE-M21
Status RUNNING

0

SOFTWARE

Why a Continuous-Time Network Rather Than an Alarm

An alarm that fires when moisture drops below a fixed number is a much smaller project than this one, and it would work. So the network has to justify itself.

The first justification is time, and it is the real one. This model is told how much actual time passed between readings, and its state evolves accordingly. A slow slide and a sudden drop produce different internal trajectories even when they end at the same value. That is a property a threshold cannot have, and it matters because the subject is an environment, where almost everything that matters is a shape over hours rather than a value at an instant.

The second is that several senses arrive together and interact. Drought in the soil genome is not moisture alone — it is moisture falling while temperature rises while light has been high for hours, three departures reinforcing each other. A network integrating all of them at once handles that without me writing a rule for every combination, and a single drifting sensor carries less weight when nothing else agrees with it.

The third is continuity. The state carries forward, so the object has a recent past. It does not evaluate each reading from a blank start.

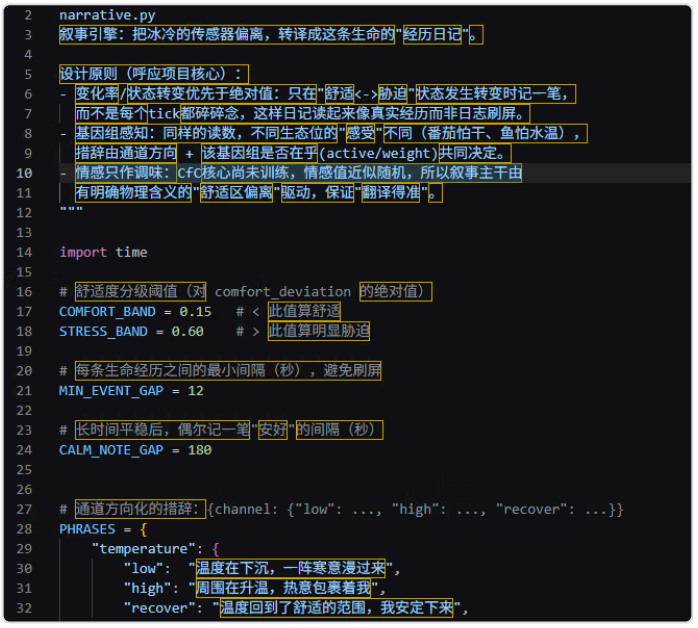
Now the correction, which is why this card was rewritten. An earlier version of this card claimed that the system prioritises rate of change over absolute value. That is the design intention and it is **not currently implemented.** My own technical document is explicit: within five weeks, only instantaneous readings are used, and the rate-of-change principle appears only in the calibration logic, not in live inference. The interface for adding it exists and it is a named future step. Until it is built, the honest statement is that the network handles time *internally* through its own dynamics, not that the system explicitly reasons about rates.

Restating the limit from RE-I22: the weighting rules that aim any of this at drought are authored by me, and the network is untrained. Continuous-time dynamics are a real reason to use this architecture. They are not a licence to describe designed behaviour as emergent.

TYPE 🟨 INSIGHT
DATE 2026.7.20
ID RE-I14
Status LOAD-BEARING

0

The Network Is Not Trained — Saying So Before Anyone Asks



```
narrative.py
叙事引擎：把冰冷的传感器偏离，转译成这条生命的"经历日记"。

设计原则（呼应项目核心）：
- 变化率/状态转变优先于绝对值：只在"舒适<->胁迫"状态发生转变时记一笔，
  而不是每个tick都碎碎念，这样日记读起来像真实经历而非日志刷屏。
- 基因组感知：同样的读数，不同生态位的"感受"不同（番茄怕干、鱼怕水温），
  措辞由通道方向 + 该基因组是否在乎(active/weight)共同决定。
- 情感只作调味：CfC核心尚未训练，情感值近似随机，所以叙事主干由
  有明确物理含义的"舒适区偏离"驱动，保证"翻译得准"。
"""

import time

# 舒适度分级阈值（对 comfort_deviation 的绝对值）
COMFORT_BAND = 0.15   # < 此值算舒适
STRESS_BAND = 0.60    # > 此值算明显胁迫

# 每条生命经历之间的最小间隔（秒），避免刷屏
MIN_EVENT_GAP = 12

# 长时间平稳后，偶尔记一笔"安好"的间隔（秒）
CALM_NOTE_GAP = 180


# 通道方向化的措辞：{channel: {"low": ..., "high": ..., "recover": ...}}
PHRASES = {
    "temperature": {
        "low":  "温度在下沉，一阵寒意漫过来",
        "high": "周围在升温，热意包裹着我",
        "recover": "温度回到了舒适的范围，我安定下来",
```

This is the card I would most like to be judged on, and it is the one that makes the project smaller.

The emotion network in this device has never been trained. Its internal weights are random. It runs in inference mode and no learning happens while it is running.

This is not an accident and it is not a bug. It is the same condition as the wild-type entity in Investigate — a randomly initialised core with no pre-training — and it is deliberate. But it has a consequence I have to state plainly rather than let a viewer infer wrongly: the five emotion values coming out of this network are not meaningful in themselves. They are a fixed but arbitrary function of the input. Valence is not a judgement about whether things are good. It is a number produced by untrained weights.

I did not decide this by reasoning about it in the abstract. I found it while writing the narrative engine — the part that turns readings into a written diary — and discovering that if I drove the diary from the emotion values, the entries were nonsense. So I wrote in the comments at the top of that file that the emotion values are approximately random and that the diary would instead be driven by the comfort-zone departures, which have unambiguous physical meaning. That comment is the most useful sentence in my codebase.

So here is the honest account of what makes the light change colour.

The chain that carries meaning is: sensor reading, comfort curve, genome weight, and then straight into either the diary or the light. Every step in that chain is a rule I wrote, and every step is defensible,

because a departure from a comfortable moisture band genuinely does mean the soil is drying.

The network's contribution is different in kind. Because it runs in continuous time and carries its state forward, it contributes how the response moves — how quickly the state shifts when several senses change together, how it lags, how it settles afterwards. That is a real temporal texture and it is why the light does not snap between states like a threshold alarm. It is not knowledge about ecology.

Which gives the answer to RE-Q6. When the light breathes and someone says it feels alive, the correct response is: the colour is designed, the timing is dynamics, and neither is understanding. If a drought demonstration looks convincing because I wrote a moisture weight of 0.9 and mapped negative valence to red, that is successful design and I should call it design.

The uncomfortable version, stated once so it cannot be said I avoided it: with the network untrained, a large part of what this device does could be achieved without it. The network earns its place through temporal behaviour and through architectural continuity with Investigate — not because it knows anything. Training it on real long-term deployment data is the obvious next step and it is outside five weeks.

TYPE 🟨 INSIGHT

DATE 2026.7.20
ID RE-I22
Status LOAD-BEARING

0

How to Read the Software Section

Three sentences, so this section is not read as a list of features.

Everything here is a rule I wrote. The hardware section was about what the object can physically feel. This section is about what it does with that — which senses matter, what counts as unusual, and how a feeling becomes light. None of it was discovered by the machine. All of it is authored, and the point of writing it out is that authored rules can be argued with.

The order matters. Read it as: the world arrives as numbers, the numbers become a departure from normal, the departure gets weighted by a nature, the weighted signal goes into a network, the network's state becomes light. Each of those arrows is a card, and each arrow is a place where I made a decision.

The most important card in this section is RE-I22, and it is the one that says what the network is not doing. If you only read one card here, read that one.

TYPE ⬜ CONTEXT

DATE 2026.7.1 ID RE-S4 Status SUPPORTING

0

A Genome Is a Menu of What to Care About

In Investigate, genome meant temperament — eight numbers describing an individual's innate leanings. In Realise I kept the word and gave it a different job, which is confusing on purpose and worth defining tightly.

Here a genome is a text file. For each of the senses the body has, it says three things: how much this nature cares about that sense, what range counts as comfortable, and how far outside that range counts as real distress. That is all. You can open it in any text editor and read it, and if you disagree with a number you can change it and run the thing again.

What this is not. It is not a different circuit board for each niche. It is not a personality for you to get attached to. And it is emphatically not evidence that the network understands tomato biology — it is evidence that I read about tomatoes and wrote some numbers down.

What it preserves from Investigate is the structural idea that one core can be biased into different dispositions by a small set of parameters. What changed is the direction of the bias: in Investigate the disposition was social, and here it is ecological.

The engineering test is blunt. If switching from soil to water requires me to unsolder anything, then the chameleon claim in RE-Q3 is false and I have quietly built a product line instead of one body. It does not, and the code that does the switching is in RE-M25 — although whether the object is allowed to switch turns out to be a separate question, and I answered it differently. See RE-I32.

TYPE 🟨 INSIGHT

DATE 2026.7.1
ID RE-I11
Status LOAD-BEARING

0

Two Layers Before the Network Sees Anything

A sensor reading is not a feeling, and the gap between them is where most of my design decisions live. There are exactly two layers in that gap, and keeping them separate is deliberate.

Layer one turns a number into a departure. Each sense has a comfortable band and, further out, a distress threshold. Inside the band, the output is zero — nothing to report. Outside it, the output rises smoothly toward one as the reading approaches distress, and stops at one so that an absurd reading cannot overwhelm everything else. Crucially the output is signed: too little is negative, too much is positive. Soil that is bone dry and soil that is waterlogged are both far from comfortable, and they are opposite emergencies, so collapsing them into a single "badness" score would throw away the only thing that tells them apart.

Layer two multiplies that departure by how much this nature cares. A departure of 0.8 on a channel weighted 0.9 arrives as 0.72. The same departure on a channel weighted 0.15 arrives as 0.12. Nothing is hidden; things are simply prioritised.

Why two layers rather than one threshold. Because they answer different questions and I want to be able to change one without the other. Layer one asks what is unusual for this particular place — and the numbers in it can eventually come from observing the actual environment rather than from my reading, which is RE-M26. Layer two asks what this kind of life prioritises, and

```
"""
sensory_mapper.py
两层感官加权变换：舒适度曲线 -> 基因组权重
把原始传感器读数转成喂给CfC核心的加权偏离向量。

不依赖任何机器学习库，纯Python+可选numpy，可以脱离硬件独立测试。
"""

import json
import os
import logging

logger = logging.getLogger("sensory_mapper")

# 固定的通道顺序，任何地方引用这个向量都按这个顺序
CHANNEL_ORDER = [
    "temperature", "moisture", "voc", "humidity_air",
    "light", "sound", "ph", "ec"
]


def comfort_deviation(value, lo, hi, stress_lo, stress_hi):
    """
    返回有符号偏离度，范围裁剪到 [-1, 1]。
    0 = 完全舒适；负值 = 低于舒适区(越负越严重)；正值 = 高于舒适区(越正越严重)。
    """
    if lo <= value <= hi:
        return 0.0
    elif value < lo:
        if value <= stress_lo:
            return -1.0
        tress_lo 导致除零
```

Padlet Drive

sensory_mapper

that is a design claim about ecology that stays fixed. Drought in soil and poor ventilation in a room are different signatures even though temperature appears in both stories, and the two-layer split is what lets that be true.

One detail with a principle inside it: a sense that is switched off for a niche — moisture in an air genome, since there is no substrate — is skipped entirely rather than being given a weight of zero. Those are different statements. Zero means this nature does not care. Switched off means this question has no meaning here. The code distinguishes them because the difference is real, and the raw reading is written to the log either way.

Both layers are design. Neither is learned.
That distinction is the whole subject of RE-I22.

TYPE 🟨 INSIGHT

DATE 2026.7.1
ID RE-I12
Status LOAD-BEARING

ATTACH： 挂 感官加权系统技术设计文档.md

0

Building the Weighting Layer — And What It Does When a Sensor Lies

Writing RE-I12 as running code took one file, and most of the work in it is not the arithmetic. The arithmetic is four lines. The rest is deciding what to do when the world does not cooperate.

Four situations, each with a decision behind it.

A sense is switched off for this niche. Skip it entirely, output zero, but still write the raw reading to the log. Logging a channel nobody is listening to sounds pointless until you consider the case where the wrong genome is loaded — a soil genome in a fish tank. If the muted channels were never recorded, there would be no way to ever notice. The body senses fully; the genome only decides what it feels.

A reading is missing because the sensor is not connected. Output zero, meaning neutral, and write a warning into the log. Neutral rather than distressed, because an absent sensor is not evidence of a problem — the whole argument in RE-M12 and RE-M18 is that absence must stay absence.

A reading arrives that is physically impossible, like a pH of fifteen on a scale that ends at fourteen. Treat it as a broken sensor rather than an emergency. Neutral, warning, move on. This one is the least intuitive and the most important: the naive behaviour is to feed it through the curve, get maximum distress, and turn the light red. A device that panics loudest when its sensors are broken is worse than useless, because it is most dramatic exactly when it is least informed.

A reading is not a number at all, because the line arrived garbled. Same treatment.

What these four have in common is that the system's response to not knowing is to say nothing rather than to guess. That is the same commitment as RE-M12 at the software end, and it is the reason I can claim in RE-I30 that missing senses stay missing in the story.

TYPE 🟩 MAKING

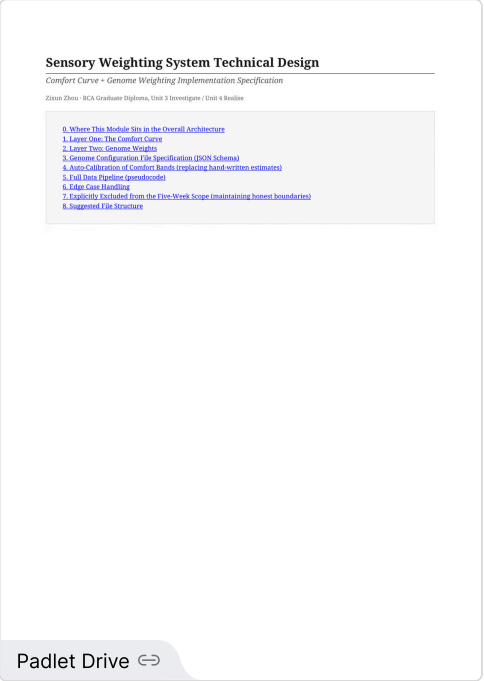
DATE 2026.7.1
ID RE-M24
Status LOAD-BEARING

ATTACH： sensory_mapper.py。

0

Sensory Weighting System Technical Design
technical support & generated by claude sonnet 5

0



Sensory_Weighting_System_Technical_Design.pdf

Seven Universal Sensory Channels

One shared body of senses. Genomes differ in *care*, not in whether the channel exists.
Seven channels: **T** temperature (medium) · **M** moisture (substrate water; weight forced to zero when physically N/A) · **G** air & VOC (breath, staleness, ventilation) · **L** light (day/night rhythm) · **S** sound (disturbance, flow, noise) · **pH** acidity (wet media) · **EC** conductivity / nutrients (starvation vs salt stress).
Principle: a channel weighted to zero is still recorded. The body senses fully; the genome only decides what it feels.
That leaves room later for “this looks more like water — switch genome?” without having been blind (`RE-Q3`, `RE-I11`).
Hardware on the bench may lag the ideal list (e.g. photoresistors before BH1750, BMP280 before BME680, pH paused). Those gaps belong on MAKING cards as RUNNING — they do not rewrite this principle.

TYPE 🟨 INSIGHT
DATE week 5
ID RE-I10
Status LOAD-BEARING

0

From Feeling to Light

Five dimensions come out of the network. Not all five reach the light, and I would rather say which than imply a completeness the code does not have.

Valence becomes colour. Distress runs toward amber and red; ease runs toward green and cyan. In the driver this is not a continuous rainbow but five named states — settled, calm, alert, uncomfortable, threatened — with the colour sliding smoothly between them rather than snapping, because a hard jump reads as a notification and a slide reads as a body changing.

Arousal becomes breathing speed. The rhythm rises and falls continuously; higher arousal breathes faster. In practice the driver deliberately compresses this into a narrow, slow range — the fastest it will go is still slower than a person's resting breath. A fast blinking object is an appliance demanding attention. This one is not allowed to demand.

Safety becomes steadiness. Below a threshold, a small random unevenness is added to the brightness. Not a flicker, not an alarm — the difference between a steady hand and a slightly unsteady one.

Seeking raises the pulse rate at extremes, and a tone is reserved for extremes only. Never speech. That is argued separately in RE-I21.

Connection is currently not mapped to anything.

The expression code uses valence, arousal, safety and seeking; connection is computed and unused. I am stating that rather than describing an intended flowing-ring behaviour as if it existed.

And there is a specific reason connection is the hard one, which is worth more than the missing feature. In an earlier version, connection was going to respond when a person came near. That is the motion sensor from RE-I20 reappearing in software after I had physically removed it from the board — the same wrong idea, wearing code instead of a component. The test in RE-Q2 rules it out: does the output report the environment, or audition for affection. So connection has to mean the *sensed medium* answering back, and I have not yet found a version of that which is honest rather than decorative. It stays unmapped until I do.

One number I should not overstate: a specific breathing period appears in my portfolio deck as a calm resting rhythm. That figure was chosen for comfortable human peripheral reading and then adjusted again in the driver. It was not discovered by the network, and the deck and the code should be reconciled rather than quoted at different values.

TYPE 🟨 INSIGHT
DATE week 1
ID RE-I13
Status LOAD-BEARING

0

```
"""
expression_mapper.py
把五维情感状态转成物理表达参数(LED颜色+呼吸节奏)。
硬件端还没有RGB灯环时，这里先输出数值+模拟打印，
接线完成后直接把 rgb 三元组发给 Arduino 控制 WS2812 即可，逻辑不用改。
"""


def emotion_to_expression(emotion: dict) -> dict:
    """
    emotion: {"valence":..., "arousal":..., "safety":..., "connection":..., "seek
             每个值范围约 [-1, 1] (来自life_core.py的tanh输出)
    返回: {"rgb": (r,g,b), "breathe_hz": float, "steadiness": float}
    """
    valence = emotion["valence"]
    arousal = emotion["arousal"]
    safety = emotion["safety"]
    seeking = emotion["seeking"]

    # valence -> 色相: 负向偏红橙，正向偏青绿
    if valence >= 0:
        r = int(255 * (1 - valence) * 0.5)
        g = int(180 + 75 * valence)
        b = int(150 + 60 * valence)
    else:
        r = int(180 + 75 * (-valence))
        g = int(120 * (1 + valence))
        b = int(90 * (1 + valence))
    r, g, b = (max(0, min(255, c)) for c in (r, g, b))

Padlet Drive
频率: -1时约0.2Hz(缓慢), +1时约2Hz(急促)
 + (arousal + 1) / 2 * 1.8
```

expression_mapper

The Calibration Window —
Letting It Watch Before It Judges

RE-Q4 asks where the comfortable ranges come from. Right now the honest answer is: I estimated them from reading, and every genome file admits this in a field near the top that says so in plain text. I have left that field in rather than deleting it.

This card is the mechanism that replaces the estimate, and it is written and tested but not yet run on real long-term data.

How it works. When the device is placed somewhere new, it spends the first several days making no judgements at all. It only records. Then it takes each sense's recorded history, finds the average and the spread, and defines comfortable as within one spread of the average and distressed as beyond three spreads. It writes those numbers back into the genome file and changes the status field from "estimated" to "calibrated from real observation."

Three details with reasoning attached.

It refuses to calibrate on too little data, and says so rather than producing a confident band from six readings.

It does not calibrate senses that are switched off for this niche, since there is nothing meaningful to observe.

And during the observation period the light holds a soft neutral white — not because that is pretty, but because it is the honest expression of the state the device is actually in. It is not calm. It is not distressed. It has no basis yet for either, and the light should say that rather than defaulting to a comfortable green.

Why this is worth the complexity. It moves the answer to "where did these thresholds come from" from I read some papers to the device measured this specific place for a week. Those are very different claims to defend. It does not make the thresholds objective — a week of observation in an already-struggling environment would calibrate distress as normal, which is a real weakness and the reason the estimated values remain as a fallback. What it does is make the origin of every number local and inspectable: you can always ask which days, under what conditions, with which genome.

And it happens to be the same shape as something already in the project's narrative from Investigate — a new thing spends its earliest period simply taking in its surroundings before it has any basis for judgement. I did not design the mechanism to fit the metaphor. I noticed afterwards that the honest engineering solution and the developmental story were the same shape, which is the kind of coincidence worth pointing at rather than pretending was planned.

TYPE 🟩 MAKING

DATE 2026.7.1

```python
"""
calibration.py
用部署头几天的真实观测数据，自动校准每个通道的舒适区间，
替代genome JSON里手写的文献估计值。
"""

import json
import numpy as np
import logging

logger = logging.getLogger("calibration")


def calibrate_comfort_band(readings, margin_std=1.0, stress_std=3.0):
    """
    readings: 某个通道过去N天的原始读数列表(list of float)
    返回该通道校准后的 (lo, hi, stress_lo, stress_hi)
    """
    if len(readings) < 10:
        raise ValueError("样本数太少，无法可靠校准，至少需要10个读数点")
    arr = np.array(readings, dtype=float)
    mean = float(np.mean(arr))
    std = float(np.std(arr))
    if std < 1e-6:
        # 极端情况：这段时间读数几乎不变，给一个最小容差避免舒适区宽度为0
        std = abs(mean) * 0.05 + 1e-3
    lo = mean - margin_std * std
    hi = mean + margin_std * std
    stress_lo = mean - stress_std * std
    stress_hi = mean + stress_std * std
                tress_lo, stress_hi
```

Padlet Drive

calibration

ID RE-M26
Status RUNNING

0

Moving the Emotion Core Into the Object

The network doing the feeling is the same one built in Investigate, moved across without retraining. That is the point rather than a shortcut, and RE-S3 lists it as the main thing that travels from AL- to RE-.

What it actually is: a CfC network wired by AutoNCP. Both of those names describe a neural network that runs in continuous time — meaning it is told how much real time has passed between one reading and the next, and its internal state evolves accordingly. A conventional network sees a sequence of steps and has no idea whether they were two seconds or two hours apart. This one is told, every time.

That is the whole reason it is here rather than a simpler thing. This object's subject matter is a slow environment. A drop in soil moisture over four hours and the same drop over four minutes are different events, and only a model that knows about elapsed time can treat them differently.

The specific shape: eight weighted senses go in, twenty-eight neurons in the middle, five values come out — the same five-dimensional emotion space from Investigate. Valence, arousal, safety, connection, seeking.

Two implementation details that carry meaning.

The internal state persists between readings. Each tick does not start from blank; it continues from where the last one left off. That continuity is the closest thing in this system to the object being continuously alive rather than repeatedly consulted, and it is the same property that made the wild-type experiment's crash-and-restart seams matter so much in Investigate.

And each genome nudges the starting state slightly, using one of the temperament parameters carried over from the eight-parameter system. Deliberately gently — the offset is scaled right down, because a large one would mean the genome was dictating the output rather than tilting it.

What did not travel: the caregiver language model, the simulated apartment, and any suggestion that five weeks of this equals long ecological learning.

TYPE 🟩 MAKING

DATE 2026.7.1
ID RE-M23 Status
LOAD-BEARING

```
"""
life_core.py
把 SensoryMapper 输出的加权偏离向量，喂入 CfC + AutoNCP 网络，
输出五维情感状态 (valence, arousal, safety, connection, seeking)。

这是我们上次确认过的架构：CfC核心完全不知道"基因组"的存在，
它只是在处理一个每次基因组切换后会变化的输入向量。
基因组的baseline_params(arousal_baseline等)作为初始隐藏状态偏置注入，
不需要重新训练网络。
"""

import torch
import torch.nn as nn
from ncps.torch import CfC
from ncps.wirings import AutoNCP

from sensory_mapper import SensoryMapper, CHANNEL_ORDER

EMOTION_DIMS = ["valence", "arousal", "safety", "connection", "seeking"]


class EmotionCore(nn.Module):
    """
    输入：8维加权感官向量(来自SensoryMapper)
    输出：5维情感状态(来自AutoNCP的motor neurons)
    """

    def __init__(self, input_size=8, n_neurons=28, n_outputs=5):
        super().__init__()
        wiring = AutoNCP(n_neurons, n_outputs)
        self.cfc = CfC(input_size, wiring)
        self.state_size = wiring.units
```

Padlet Drive

life_core

0

Genome Presets & Sensor Inference Design

technical support & generated by claude sonnet 5

Genome_Presets_and_Sensor_Inference_Design.pd

A Genome Is a Sensory Weighting Table, Not Different Hardware

In Investigate, genome meant temperament — innate bias for an individual entity. In Realise, the same word is overloaded on purpose, then redefined tightly:
A genome is a **JSON weighting table**: per channel, a weight (0–1), a comfort range, and a sensitivity curve. Switching file is how one body wears soil/root, freshwater, or indoor air (`RE-Q3`).
What this is not:
- not a different PCB per niche
- not a personality for the user to bond with
- not proof that the CfC "knows" tomato biology

What this preserves from Investigate: the idea that the *same core* can be biased into different dispositions by a compact parameter set. What changes: the disposition is ecological, not social.
Engineering test: if I cannot change niche without resoldering, the chameleon claim is false.
TYPE 🟨 INSIGHT
DATE week 5
ID RE-I11
Status LOAD-BEARING

0

Two-Layer Sensory Weighting — Comfort Curve × Genome Weight

Raw sensor numbers are not feelings. Realise inserts two authored layers before the CfC sees the world:
1. **Comfort curve** — how far a reading sits from a local comfort band (preferably calibrated from early deployment observation, not only from a paper — `RE-Q4`).
2. **Genome weight** — how much this nature cares about that channel right now.

The product fed to the network is the combination: a weighted stress / relief signal per channel (and jointly across channels for signature events).
Why two layers rather than one magic threshold: it keeps "what is unusual for this place" separable from "what this niche prioritises." Drought for soil is not the same signature as poor ventilation for indoor air, even if temperature appears in both stories.
These layers are **design**. They must stay architecturally distinct from the CfC's continuous-time dynamics (`RE-Q6`, `RE-I22`). Full technical write-up hangs as supporting document (`RE-S4` / sensory weighting PDF).

TYPE 🟨 INSIGHT
DATE week 1
ID RE-I12
Status LOAD-BEARING

0

The Soil Genome — Deciding What This Nature Panics About First

A genome in Realise is not a different machine. It is a text file — a JSON, meaning a plain list of settings a human can open and read — that says, for each of the senses the body has, how much this particular nature cares about it. Same egg, same probes, same emotion network. Different priorities.

Writing the soil one meant asking a question I could not answer from a datasheet: if this were a tomato root, what would it notice first?

Moisture came out top, and not by a small margin. I gave it the highest weight in the file, 0.9 out of 1. A root's emergency is thirst; everything else is context. Temperature next at 0.8, because roots cook and roots freeze and both are fast. Light at 0.7, which surprised me until I thought about it properly: light is not directly a root's problem, but a long bright hot stretch is what makes the ground dry out, so light is early warning rather than a threat in itself.

Then the ones I deliberately turned down. Sound sits at 0.15 — a root is not indifferent to vibration, but a noisy room is not a soil emergency, and if I had weighted it higher the object would have started reacting to me walking past, which is the pet brief crawling back in through the numbers. Air staleness at 0.6, kept in as a hint about rot rather than a main signal.

The event I want to be able to demonstrate is a combination rather than a single reading: the ground dries, the temperature climbs, and it stays bright for hours. Three independent senses agreeing. I built it that way because a single sensor drifting is usually just a sensor drifting — one reading can lie, three readings agreeing rarely do. That is also the reason the CfC network is still in this project instead of a simple threshold alarm, and it is argued properly in RE-I14.

What I have to say plainly: the actual numbers in this file are still estimated from literature, not measured in a real pot of soil. The file says so itself — there is a field near the top called _calibration_status and it currently reads "using literature-estimated defaults, not yet field-calibrated". I left that line in rather than deleting it, because RE-Q4 promises that where the thresholds come from is inspectable, and a promise you delete from the file is not a promise.

TYPE 🟩 MAKING

DATE 2026.7.1
ID RE-M2
Status RUNNING

0

---

he Water Genome — Reasoning Around a Sense I Do Not Have

The freshwater genome ran into a wall the soil one did not, and the wall turned out to be the interesting part.

The thing that actually kills a pond or a tank is oxygen dropping. I do not own a dissolved oxygen probe, and they are expensive. So the first version of this file was a lie waiting to happen: I could easily have written a number called "oxygen" and computed it from other readings, and nobody looking at a demonstration would have known.

I considered three responses honestly. Fake it in software and never mention it — no. Buy the proper probe in week one — deferred rather than abandoned, because the money and the calibration time were not there. Or admit the gap and change what kind of claim the device is making about water.

I took the third, and it changed the design rather than just the caption. Instead of asserting a state, the water genome watches for a combination that is suspicious: water temperature climbing, at high weight, 0.9 — warm water holds less oxygen, so temperature is the closest honest proxy I have — alongside pH

drifting at 0.8 and conductivity moving at 0.7. Any one of those alone means very little. Together, in that pattern, they mean something is going wrong in a way that is consistent with oxygen falling.

Which gives the three natures three genuinely different relationships to their own certainty. Soil can ask the ground a direct question and get a direct answer: are you dry. Air, if the right chip is fitted, can ask directly whether it is stale. Water cannot ask its most important question at all, and has to reason sideways from things it can measure.

And I have to go further, because there is a second gap and it is worse. The two channels this genome leans on after temperature are pH and conductivity — and the pH probe is bought but still not wired (see RE-I30), while the conductivity probe is running on the manufacturer's default formula with no calibration. So the freshwater genome as written cannot be honestly demonstrated yet. It is the design, not the evidence. It is marked RUNNING for that reason and it will stay RUNNING until those two probes are real.

That distinction — I designed this versus I demonstrated this — is the same line RE-Q6 draws between designed rules and network dynamics. It is the line this whole project is trying not to blur.

TYPE 🟩 MAKING

DATE 2026.7.1
ID RE-M3
Status RUNNING

0

The Indoor Air Genome — The Room's Breath, Not Your Arrival

The third genome is the one where the old companion brief tried hardest to come back, and it is also the one with the biggest hole in it.

Indoor air is the niche closest to a person. The soil version sits in a pot; the water version sits in a tank; this one sits in a room with you in it. So every design instinct pulled toward making it react to you — brighten when you come home, settle when you sit down. I want that instinct on the record because it was strong and it was wrong. RE-Q2 asks whose feelings are being shown, and the answer for this genome has to be the room's, not yours. The air in a badly ventilated room is getting worse whether or not you have just walked in, and the entire point is to make that visible.

So the weights follow the room. VOC — chemical staleness in the air — is the highest at 0.9. Humidity and sound both at 0.6, because a room that is damp and echoing tells you something about ventilation. Temperature at 0.6. Moisture, pH and conductivity are all switched off entirely, weight zero, marked inactive — there is no soil and no water here for them to read.

That last bit matters more than it sounds. A channel at zero is still recorded. The sensor keeps sensing; the genome only decides that this nature does not care. That is what makes it possible later for the system to notice "these readings look like water, not air — perhaps the wrong genome is loaded" rather than being blind to it. RE-Q3 and RE-I11 argue that properly.

Now the hole, which I would rather write myself than have someone find. VOC is the channel this genome cares about most, at 0.9 — and there is no VOC sensor on the device. I have a BMP280, which reports pressure and temperature and cannot smell anything. The BME680 that would do it has not been bought. So the indoor air genome is currently a nature whose most important sense does not exist.

Which leaves the five-week window with an honest and slightly uncomfortable summary: I designed three natures and only one of them — soil — can actually be demonstrated right now, because it is the only one whose top-weighted channel is both wired and calibrated. Rather than quietly demonstrating all three on simulated data, the plan is to demonstrate soil properly and present the other two as designs with their missing organs named. RE-Q5 already committed me to that: five weeks can show translation under induced stress, not mastery of ecology. This is what that commitment costs when it is actually enforced.

TYPE 🟩 MAKING

DATE 2026.7.1
ID RE-M4
Status RUNNING

0

No Speech — Even Though It Would Be Easy

The object has a diary. It writes sentences like "the soil is drying out; I begin to feel thirsty." Those sentences are in the phone journal, and they are never spoken and never displayed on the object itself.

This is a restriction I imposed while holding the means to break it. The whole Investigate project used a language model. Making this thing talk would take an afternoon and it would be the single most impressive-seeming addition available to me.

The reason not to is in the field work. AL-P0 through P4 asked people who care for small animals how they first notice a state change without language. Every participant answered from before language: a sound, a stillness, a smell, a posture, or something they could only call a feeling. That pre-verbal reading is the thing this project is trying to make possible for environments — and if the object says "I am thirsty," the reading is over before it starts. You have not learned to read anything. You have been told.

There is a second reason that is about the machine rather than the person. A spoken sentence claims understanding. When something says "I feel unsafe," it is asserting an inner life. Given RE-I22 — an untrained network producing approximately arbitrary emotion values — that claim would be false in a way that light simply cannot be. Light is honest by being ambiguous. It shows a state and leaves the interpreting to you, which is exactly the division of labour the field participants described doing with their animals every day.

So: a tone is permitted at extremes, where it functions as an alarm rather than a statement. Text lives in the journal, where it is a record you choose to open, not a voice in the room. The object itself stays wordless.

The written phrases exist in both English and Chinese, and they are fixed sentences I wrote, chosen by which sense departed and in which direction. No language model runs on this device. That is worth stating precisely, because "it writes a diary" would otherwise imply something much more impressive and much less honest than what is happening.

TYPE 🟨 INSIGHT

DATE 2026.7.20
ID RE-I21
Status LOAD-BEARING

0

```
"""
narrative.py
叙事引擎：把冰冷的传感器偏离，转译成这条生命的"经历日记"。

设计原则（呼应项目核心）：
- 变化率/状态转变优先于绝对值：只在"舒适<->胁迫"状态发生转变时记一笔，
  而不是每个tick都碎碎念，这样日记读起来像真实经历而非日志刷屏。
- 基因组感知：同样的读数，不同生态位的"感受"不同（番茄怕干、鱼怕水温），
  措辞由通道方向 + 该基因组是否在乎(active/weight)共同决定。
- 情感只作调味：CfC核心尚未训练，情感值近似随机，所以叙事主干由
  有明确物理含义的"舒适区偏离"驱动，保证"翻译得准"。
"""

import time

# 舒适度分级阈值（对 comfort_deviation 的绝对值）
COMFORT_BAND = 0.15   # < 此值算舒适
STRESS_BAND = 0.60    # > 此值算明显胁迫

# 每条生命经历之间的最小间隔（秒），避免刷屏
MIN_EVENT_GAP = 12

# 长时间平稳后，偶尔记一笔"安好"的间隔（秒）
CALM_NOTE_GAP = 180


# 通道方向化的措辞：{channel: {"low": ..., "high": ..., "recover": ...}}
PHRASES = {
                                                    "无下沉，一阵寒意漫过来",
```

Padlet Drive

narrative

The Engine Can Change Its Nature. The Object Is Not Allowed To.

This is a decision where the code and the product deliberately disagree, and I want the disagreement visible rather than tidied away.

The engine supports switching genome at runtime. One function call swaps the weighting table and the object is now reading its world as a different kind of life. It works, and RE-M25 describes it.

The product refuses to use it. In the application, a genome is chosen once, when a life begins, and never changes for as long as that life exists.

I went back and forth on this and the argument that settled it was about what the word nature is doing. If you can change what something fundamentally cares about with a dropdown menu, then it was never a nature — it was a costume, and the object is a demonstration rig wearing themes. RE-Q3 asks whether one body can wear many natures. The answer I have landed on is: one body, yes, across many lives — but not one life trying on several. A tomato does not become a fish in the afternoon.

There is also a technical fact that supports it and that I did not design for. Switching genome resets the network's internal state, because the state was accumulated under a different set of priorities and carrying it across would be meaningless. So a switch is not seamless — it is a discontinuity, a small death and rebirth. Once I noticed that, pretending it was a settings change felt dishonest.

This has a cost I am accepting knowingly. It makes the most impressive possible demonstration unavailable: same hardware, same room, flip the file, watch the light reinterpret the identical world. That demonstration would be a genuinely good argument for RE-Q3 and I have given it up.

What replaces it is in RE-P3: run the same recorded conditions through two separately born lives with different natures, and compare. Less theatrical. It shows the same thing without requiring the object to contradict what it says it is.

TYPE 🟨 INSIGHT

DATE 2026.7.20
ID RE-I32
Status LOAD-BEARING

0

Switching Genome Is Loading a File

Body:

The engineering claim behind RE-Q3 comes down to one thing: is changing niche a matter of changing a file, or a matter of changing the machine.

It is a file. The function that does it loads a different text file into the weighting layer and deliberately keeps the same network weights. Nothing is recompiled, no hardware is touched, no soldering iron is involved. The same body, the same probes, the same core — a different set of priorities.

Keeping the network weights identical across the switch is the part that matters. If a genome swap also swapped the network, then soil and water would be two different brains and the chameleon idea would be marketing. Because the core is the same, what changes is genuinely only what this life attends to, not what it is capable of.

The one thing that does reset is the accumulated internal state, for the reason given in RE-I32: a state built up under one set of priorities does not mean anything under another.

```
def switch_genome(self, genome_path):
    """
    运行时切换基因组：只替换感觉权重表(SensoryMapper)并重置隐藏状态，
    刻意保留同一个EmotionCore网络权重不变——这样切换的是"此刻在乎什么"
    （不同生态位的视角），而不是换一个完全不同的大脑，符合项目"变色龙"设计。
    """
    self.mapper = SensoryMapper(genome_path)
    baseline = self.mapper.baseline_params
    arousal_baseline = baseline.get("arousal_baseline", 0.0)
    self._init_hidden(arousal_baseline)
```

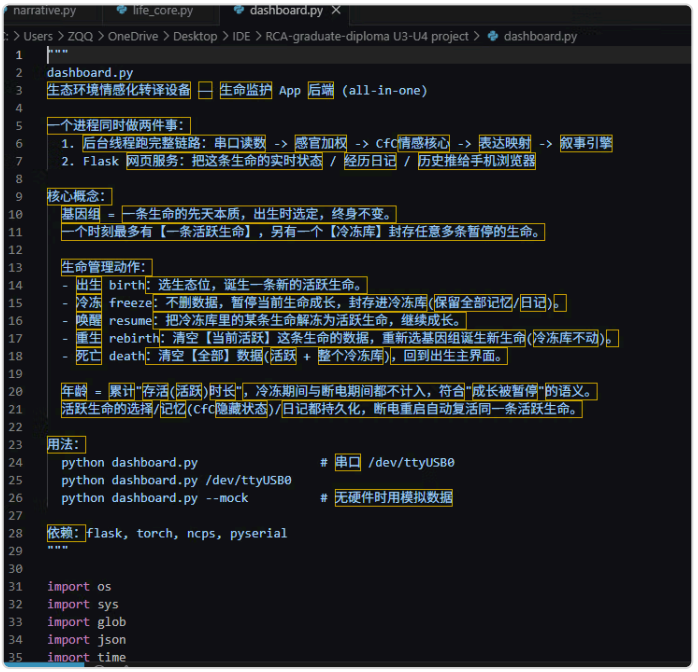
So the mechanism exists, works, and is verified. And the product then declines to expose it as a live switch, which is RE-I32. Both of those are true at once, and I would rather this Padlet show a capability that was built and then deliberately restrained than imply the restraint was a limitation.

TYPE 🟩 MAKING

DATE 2026.7.20
ID RE-M25
Status DONE

0

Birth, Freezing, Rebirth, Death — Building a Life You Can Lose



The application around all of this is not a settings panel. It is organised around one active life at a time, and around a small set of actions that are deliberately weighty.

A life is born: you choose its nature, and from that moment it accumulates a history. It can be frozen, which suspends it without deleting anything — every diary entry and every scrap of accumulated state is preserved, and it goes into storage while another life becomes active. It can be woken again later and carries on. It can be reborn, which erases this particular life and starts a new one. Or there is death, which clears everything, including storage, and returns to an empty first screen.

Age counts only time spent actively alive. Frozen time does not count. Time while the device is switched off does not count. That was the first decision in this design and everything else followed from it, because it meant age had to mean accumulated experience rather than elapsed calendar time — which is the only definition consistent with what Investigate was arguing about persistence and continuity.

The state persists across a power cut and comes back as the same life. That matters more than it sounds: in Investigate, the wild-type run had crash-and-restart seams that had to be disclosed in every card that touched them, because a discontinuity in the entity's history is a discontinuity in the claim. Building the persistence properly here is that lesson paid forward.

Two things I want to say honestly about it.

This is the part of the project that most risks pulling back toward the companion object I rejected in RE-I1. Freezing and death are emotionally loaded actions. A person who freezes a life rather than deleting it is treating it as something that can be harmed, and my field participants used exactly that vocabulary — care, responsibility, the fear of damaging something once it has become singular.

But I do not think this is the pet brief returning, and the test in RE-Q2 is why. None of these actions makes the object respond to you. They govern whether it is reading its environment at all. The weight is not affection, it is the difference between an instrument you can reset without consequence and a record that means something because it could be lost. RE-Q4's calibration argument depends on exactly that: a device whose history can be casually wiped has no basis for knowing what normal looks like here.

TYPE 🟩 MAKING

DATE 2026.7.20
ID RE-M32
Status LOAD-BEARING

0

Driving Thirty LEDs Without the Convenient Library

The light ring is thirty individually addressable LEDs. Each one wants its colour delivered as a precisely timed sequence of electrical pulses, at a rate where being slightly late corrupts the message — and a Raspberry Pi runs a normal operating system that can interrupt whatever it is doing at any moment. The usual answer is a well-known library that handles this. Installing it dragged in a stack of dependencies I did not want on a device meant to boot reliably inside a sealed shell.

So the driver instead borrows a different piece of hardware. The Pi has an interface built for talking to chips at a fixed clock rate, and the driver uses it to fake the LED protocol: every single bit of colour data is expressed as three bits on that interface, one pattern meaning one and another meaning zero. The timing then comes from dedicated hardware rather than from software hoping to be scheduled on time. The whole driver is about forty lines and has no dependencies beyond one standard module.

Above it sits the expression loop, running fifty times a second in its own thread. It reads the current state, decides which of the five moods applies, slides the colour gradually toward that mood's colour, and modulates brightness as a slow sine wave whose frequency comes from arousal. Threatened states get a raised minimum brightness so distress is visible across a room rather than only when you are standing over it. Low safety adds the small unevenness described in RE-I13.

Two decisions I want on the record.

Every drawing operation is wrapped so that a failure cannot take down the main program. If the light dies, the sensing and the reasoning continue and the logs keep being written. The expression layer is allowed to fail alone.

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And when nothing is alive — before a life has been started — the ring holds a very dim, very slow blue. Not off. Off would be indistinguishable from unplugged, and the object's entire claim is that its state is readable from across the room. Dormant has to look like dormant.

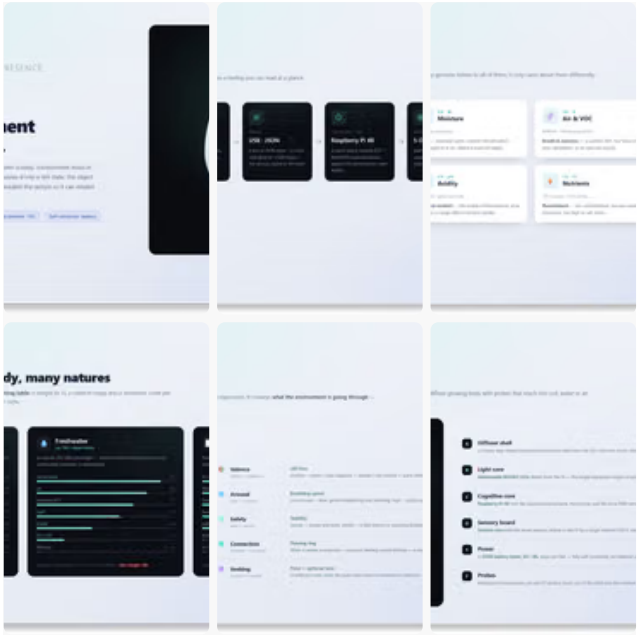
TYPE 🟩 MAKING

DATE 2026.7.20
ID RE-M27
Status LOAD-BEARING

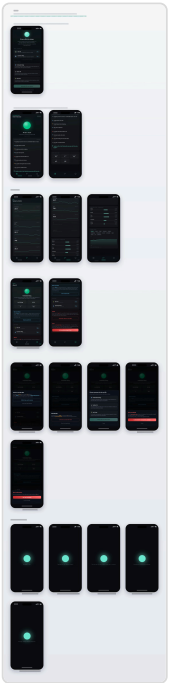
0

APPLICATION

0



The Phone Page Is a Journal, Not a Cockpit



There is a phone interface. It is a small web page served by the device itself over the local network, installable to a home screen and able to open without a connection. RE-I7 already committed to it being secondary, and this card is where that commitment had to survive contact with actually building it.

It shows the current readings, the current state, the diary, and the life-management actions from RE-M32.

The temptation was immediate and strong. Once the data is on a screen, charts are easy, and charts are satisfying, and a well-made dashboard would be the fastest way to make this project look finished. I have watched myself want to add a graph of moisture over time roughly once a week.

The rule that keeps it in its place: nothing may appear on the phone that is required to understand the object. The phone can tell you more, and more precisely, and further back in time. It cannot be the only place a state is legible. If I ever find myself needing to open it to know how

things are going, RE-Q1 has failed and the failure is in the light, not in the phone.

There is a structural reason it stays honest, too. The reasoning process writes its current state to a small file, and the web page reads that file. The page cannot ask the network anything, cannot influence it, and does not contain any of the interpretation. It is a window onto a process that would run identically with nobody watching — which is the correct relationship for a device whose subject is an environment that does not care whether you are looking.

Where it genuinely earns its place is the diary. RE-I21 keeps language off the object; this is where language is allowed to live, as a record you choose to open. Reading back through three days of entries is a different act from glancing at a light, and it is the one thing here a chart could not do better.

TYPE 🟩 MAKING

DATE 2026.7.20
ID RE-M28
Status SUPPORTING

0

FINAL SHOW

Everything at Once — The First Full Loop

The smoke test is the moment where the two halves of this Padlet meet: real sensors on one side, all the authored rules on the other.

One command, one genome, and the whole chain runs continuously. Real readings arrive from the Arduino every two seconds. Each one passes through the comfort curves and the genome weights. The weighted vector goes into the network, which is told how much real time has passed. Five values come out. Those become a colour and a rhythm on thirty LEDs, and any transition between comfort and stress becomes a diary entry.

What it verifies is only that the chain is unbroken and does not fall over. That is not nothing — it is the first time the sequence exists as one continuous thing rather than as parts I have tested separately — but it is much less than it looks. Read it against RE-S5 for what the object can actually feel: two channels are properly real, three are rough, one is absent, one is unwired. Read it against RE-I22 for what the network is actually contributing.

The things that made it not work the first time were, without exception, boring. A speed mismatch. A field name typed differently at each end. The absent pH field arriving as nothing where the code expected something. None of them were interesting and all of them were mine.

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And one thing I should say about what this is not.
Everything up to here proves the machine works.
Nothing up to here shows that a person can read it.
That is a different question, it is the question RE-Q1
actually asks, and it belongs to the demonstration cards.

TYPE 🟩 MAKING

DATE 2026.7.20
ID RE-M29
Status RUNNING

0

土壤组

0

空气组1

0

0